

Arithmetic classification and non-Pisot singularity for Bernoulli convolutions

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Abstract

We give an arithmetic classification of singular and absolutely continuous unbiased Bernoulli convolutions for every parameter $\lambda \in (0, 1)$. The criterion is expressed through one-sided approximation by explicitly defined finite sets of algebraic units; it is an infinite approximation condition, not a finite membership algorithm. We also establish singular examples beyond reciprocals of Pisot numbers: singularity holds at the reciprocal of every quartic Salem number in $(1, 2)$ and at the reciprocal of a specified non-Pisot root of an explicit degree-31 polynomial.

1 Introduction

For $0 < \lambda < 1$, the unbiased Bernoulli convolution ν_λ is the law of

$$Y_\lambda = \sum_{n=0}^{\infty} \varepsilon_n \lambda^n, \quad \mathbb{P}(\varepsilon_n = 1) = \mathbb{P}(\varepsilon_n = -1) = \frac{1}{2}, \quad (1.1)$$

where the signs are independent. The series converges absolutely. We also use the bit form μ_λ , the law of

$$X_\lambda = \sum_{n=0}^{\infty} u_n \lambda^n, \quad \mathbb{P}(u_n = 0) = \mathbb{P}(u_n = 1) = \frac{1}{2}.$$

The relation $Y_\lambda = 2X_\lambda - (1 - \lambda)^{-1}$ preserves singularity and absolute continuity. A measure is singular if it gives full mass to a Borel set of Lebesgue measure zero. Bernoulli convolutions have pure type: they are either singular or absolutely continuous [8, Theorem 11]. The classification problem asks which of these alternatives holds for each parameter, including exceptional parameters.

For $\lambda < 1/2$, a direct interval cover proves singularity; at $\lambda = 1/2$, ν_λ is uniform on $[-2, 2]$. The interval $(1/2, 1)$ presents the arithmetic difficulty. A *Pisot number* is a real algebraic integer $\beta > 1$ whose other conjugates have modulus less than one. Erdős proved singularity when λ^{-1} is Pisot by showing that the Fourier transform does not tend to zero [3]. Garsia subsequently established absolute continuity for a different arithmetic class, including reciprocals of algebraic integers of absolute norm two whose conjugates all have modulus greater than one [4].

The almost-everywhere theory gives a complementary picture. Solomyak proved that ν_λ has an L^2 density for Lebesgue-almost every $\lambda \in (1/2, 1)$ [17]. Hochman's work relates dimension loss to very strong concentration of cylinder positions [7], and Shmerkin proved that the possible singular parameters in this interval form a set of Hausdorff dimension zero [16]. Varjú proved full dimension for every transcendental parameter in this interval [18]. These conclusions leave measure type at exceptional parameters to be resolved; full dimension alone does not imply absolute continuity. The

survey [12] describes the classical development and the roles of arithmetic, overlap, and Fourier decay.

Garsia connected singularity with entropy of finite sums [5]. Algebraic approximation also plays a central role in the dimension theory: Breuillard and Varjú showed that a dimension-deficient parameter can be approximated exceptionally closely by algebraic parameters whose Bernoulli convolutions also have dimension less than one [2, Theorem 1]. The criterion below addresses measure type and also records the number and coefficient directions of nearby algebraic targets.

A *Salem number* is a real algebraic integer $\beta > 1$ whose other conjugates consist of β^{-1} and conjugates on the unit circle. In degree four there is a single nonreal conjugate pair. Salem proved that the Fourier transform of ν_λ tends to zero whenever λ^{-1} is not Pisot [14]. Thus Fourier nondecay cannot prove singularity for Salem parameters. Marshall-Maldonado and Solomyak recently obtained quantitative decay estimates for these convolutions [11, Appendix B]; those estimates do not decide absolute continuity. The existence of singular examples with non-Pisot reciprocal is explicitly recorded as open in the recent literature [1, 9].

This paper gives an arithmetic criterion for every parameter and proves two classes of non-Pisot singularity results. The criterion in Theorem 2.1 uses finite sets of real algebraic units. Their minimal polynomials have coefficients in $\{-1, 0, 1\}$, prescribed degree, and irreducible reduction modulo two. A target unit is retained when nearby units supply sufficiently many distinct initial coefficient vectors in a fixed positive proportion of the coordinate orthants. Singularity is equivalent to one-sided approximation by these retained targets at a fixed geometric rate. All target sets are defined by finite polynomial algebra; their construction uses no information about the type of any Bernoulli convolution. The resulting classification is an approximation property, and need not provide a short test for membership at a given real parameter.

The main step behind the criterion is a realization lemma. Concentration of the measure produces many pairs of finite bit words with nearby sums and different first bits. Their difference vectors can be completed to height-one polynomials having a root strictly above the parameter. A uniform finite-order nonvanishing estimate preserves a sign change, while prime polynomials in arithmetic progressions over $\mathbb{F}_2$ supply irreducibility without altering the initial coefficients. This separates the concentration argument from the algebraic realization, and gives a way to translate overlap information into constrained algebraic approximation.

The two non-Pisot results are proved directly; the criterion then yields the corresponding approximation properties. Theorem 5.1 proves singularity for the reciprocal of every quartic Salem number in $(1, 2)$. We construct algebraic multipliers whose unit-circle conjugates become small faster than the associated cosine losses accumulate. Trace identities then localize many Fourier phases to disjoint windows of random signs. Their averages concentrate under the Bernoulli convolution while tending to zero in Lebesgue mean square. This yields singularity even though the Fourier transform tends to zero. The construction applies in particular to the two quartic polynomials specified in Corollary 5.4.

Theorem 6.1 treats a root of an explicit degree-31 polynomial with a complex-conjugate pair outside the unit circle. Here word evaluations lie in a lattice whose box volume grows slightly faster than the reciprocal parameter to the word length. A product of positive trigonometric factors has bounded Lebesgue integral but grows exponentially on typical word evaluations. The resulting saving outweighs the extra volume growth. A finite-state description of nearly cancelling frequencies controls the integral; an explicit rational calculation verifies the required moment inequalities. Appendix A provides the root enclosures, finite recurrences, and analytic error bounds for that calculation.

Sections 2–4 prove the classification and recover the basic Pisot and Garsia cases. Sections 5 and 6 establish the two non-Pisot results. All logarithms are natural unless a base is specified, and

all decimal constants used in estimates denote exact rational numbers.

2 An arithmetic criterion for the measure type

The criterion in this section is expressed through finite sets of algebraic units and the coefficient vectors of their minimal polynomials. Its proof has two parts. A sufficiently large cluster of coefficient vectors forces concentration of the Bernoulli convolution. Conversely, concentration produces many short polynomial prefixes, each of which can be completed to an irreducible polynomial with a root just above the parameter.

For integers $n \geq 1$ and $d \geq 2$, let $R(n, d)$ be the set of real algebraic units $r \in (1/2, 1)$ of degree dn whose minimal polynomial has all coefficients in $\{-1, 0, 1\}$ and is irreducible modulo 2. An algebraic unit here means an algebraic integer with algebraic-integer inverse. Write p_r for the minimal polynomial with its sign chosen so that its constant coefficient is -1 , and set

$$\pi_n(p_r) = ([t^0]p_r, \dots, [t^{n-1}]p_r) \in \{-1, 0, 1\}^n.$$

For $r \in R(n, d)$, define the finite set

$$W_n(r; d) = \{\pi_n(p_q) : q \in R(n, d), |q - r| \leq 4^{-n}\}. \quad (2.1)$$

This is a set of vectors: repeated projections, including those arising from conjugate roots of one polynomial, are counted only once.

For a bit word $u = (u_0, \dots, u_{n-1}) \in \{0, 1\}^n$, let

$$\mathcal{O}_u = \{w \in \mathbb{R}^n : (2u_i - 1)w_i \geq 0 \text{ for } 0 \leq i < n\}$$

be the associated closed orthant. A vector with a zero coordinate belongs to both choices of sign in that coordinate. For $j \geq 1$, put

$$R_*(n, d, j) = \left\{ r \in R(n, d) : \#\{u \in \{0, 1\}^n : \#(W_n(r; d) \cap \mathcal{O}_u) \geq j(2r)^n\} \geq \frac{2^n}{d} \right\}. \quad (2.2)$$

All comparisons in this definition involve algebraic numbers and finite sets. In particular, $R(n, d)$ and $R_*(n, d, j)$ can be obtained by finite enumeration of integer polynomials and exact algebraic comparisons.

Theorem 2.1 (Arithmetic criterion). *Define the parameter set*

$$\mathcal{C} = (0, 1/2) \cup \left[(1/2, 1) \cap \bigcup_{d \geq 2} \bigcap_{j \geq 1} \limsup_{n \rightarrow \infty} \bigcup_{r \in R_*(n, d, j)} [r - 4^{-n}, r) \right]. \quad (2.3)$$

For every $\lambda \in (0, 1)$, the measure ν_λ is singular if and only if $\lambda \in \mathcal{C}$, and is absolutely continuous if and only if $\lambda \notin \mathcal{C}$. Replacing $\limsup$ by $\liminf$ in (2.3) gives the same set.

Thus the degree multiplier d may be fixed before the crowding level j is chosen. For a singular parameter in $(1/2, 1)$, each level j is attained at every sufficiently large n , with the threshold in n allowed to depend on j . The predicate uses only polynomial arithmetic, one-sided approximation, and counts of coefficient vectors. Determining membership for a particular parameter can nevertheless be difficult; the theorem does not supply a finite decision procedure for arbitrary real inputs.

2.1 Measure-theoretic preliminaries

We use the bit form

$$X_\lambda = \sum_{i=0}^{\infty} u_i \lambda^i, \quad \mathbb{P}(u_i = 0) = \mathbb{P}(u_i = 1) = \frac{1}{2},$$

with law μ_λ , and write $L = (1 - \lambda)^{-1}$. The affine identity $Y_\lambda = 2X_\lambda - L$ shows that μ_λ and ν_λ have the same measure type. For a word $u \in \{0, 1\}^n$, put

$$s_u(\lambda) = \sum_{i=0}^{n-1} u_i \lambda^i. \quad (2.4)$$

Conditioned on this prefix, X_λ lies in $s_u(\lambda) + \lambda^n[0, L]$.

Lemma 2.2 (Pure type and the baseline parameters). *For $0 < \lambda < 1$, the measure μ_λ is either singular or absolutely continuous. It is singular for $\lambda < 1/2$ and uniform on $[0, 2]$ for $\lambda = 1/2$. Consequently, $\nu_{1/2}$ is uniform on $[-2, 2]$.*

Proof. Let $T_i(x) = i + \lambda x$ for $i = 0, 1$, and let $\mathcal{T}\rho = \frac{1}{2}(T_0)_*\rho + \frac{1}{2}(T_1)_*\rho$. Iteration from any probability measure ρ gives the law of

$$\sum_{i=0}^{n-1} u_i \lambda^i + \lambda^n Z,$$

where Z has law ρ and is independent of the bits. Since $\lambda^n Z \rightarrow 0$ in probability, these laws converge weakly to μ_λ . Hence μ_λ is the unique invariant probability of $\mathcal{T}$.

The operator $\mathcal{T}$ preserves absolute continuity and singularity. By uniqueness of the Lebesgue decomposition, the absolutely continuous and singular parts of its invariant probability are separately invariant. If both were nonzero, normalizing them would give two distinct invariant probabilities, a contradiction.

For $\lambda < 1/2$, the support is covered, for every n , by 2^n intervals of length $L\lambda^n$. Its Lebesgue measure is therefore zero. For $\lambda = 1/2$, the usual fair binary expansion gives the uniform law on $[0, 2]$. $\square$

We shall also use the following elementary formulation of concentration. It isolates the measure-theoretic input needed to establish membership in (2.3).

Lemma 2.3 (Compact concentration). *A Borel probability measure ρ on $\mathbb{R}$ is singular if and only if, for every $\varepsilon > 0$, there is a compact set K with*

$$\text{Leb}(K) < \varepsilon, \quad \rho(K) > 1 - \varepsilon.$$

Proof. If ρ is singular, apply inner regularity to a Borel null set of full ρ -measure. Conversely, choose such compact sets K_m with $\varepsilon = 2^{-m}$. The set $\limsup_m K_m$ has Lebesgue measure zero, since $\sum_m \text{Leb}(K_m) < \infty$, and has ρ -measure one, since every union $\bigcup_{m \geq M} K_m$ has ρ -measure one. $\square$

2.2 Algebraic clusters imply singularity

Suppose $\lambda \in (1/2, 1)$ satisfies the condition in (2.3), and fix a degree multiplier d witnessing it. For each j , there are arbitrarily large n and $r \in R_*(n, d, j)$ such that

$$r - 4^{-n} \leq \lambda < r.$$

If $w \in W_n(r; d)$, choose a polynomial p_q giving this projection. Its root q satisfies $|q - \lambda| \leq 2 \cdot 4^{-n}$. On any fixed compact subinterval of $(0, 1)$ containing λ in its interior,

$$|p'_q(t)| \leq \sum_{k \geq 1} k t^{k-1} = \frac{1}{(1-t)^2}.$$

The coefficients of index at least n contribute at most $L\lambda^n$ at λ . Thus there is a constant $B = B(\lambda)$, independent of n, d, j, w , such that for all sufficiently large n ,

$$\left| \sum_{i=0}^{n-1} w_i \lambda^i \right| \leq B\lambda^n. \quad (2.5)$$

Here we used $4^{-n} = o(\lambda^n)$.

If $w \in \mathcal{O}_u \cap \{-1, 0, 1\}^n$, then $v = u - w$ belongs to $\{0, 1\}^n$. Different w give different v . By (2.2), at least $2^n/d$ words u consequently have at least $j(2r)^n \geq j(2\lambda)^n$ neighbors v satisfying

$$|s_u(\lambda) - s_v(\lambda)| \leq B\lambda^n.$$

Partition the line into half-open bins of length λ^n , and let $\mathcal{I}$ be the bins containing at least one of these words u , where a word occupies the bin containing its prefix sum. For each $J \in \mathcal{I}$, choose one such word. Its neighbors lie in the $B\lambda^n$ -enlargement of J . Each prefix sum can belong to the enlargements of at most $N_B = 2\lceil B \rceil + 3$ bins. Counting word-bin incidences yields

$$\#\mathcal{I} j(2\lambda)^n \leq N_B 2^n, \quad \#\mathcal{I} \leq \frac{N_B \lambda^{-n}}{j}.$$

Enlarge each bin $J = [a, a + \lambda^n)$ to $[a, a + (1 + L)\lambda^n]$, and denote their union by E_j . The tail bound following (2.4) gives

$$\text{Leb}(E_j) \leq \frac{(1 + L)N_B}{j}, \quad \mu_\lambda(E_j) \geq \frac{1}{d}. \quad (2.6)$$

Absolute continuity of a finite measure makes its mass uniformly small on sets of sufficiently small Lebesgue measure. Therefore (2.6), as $j \rightarrow \infty$, rules out absolute continuity. Lemma 2.2 proves singularity.

3 Constructing the algebraic approximants

We now prove the converse in Theorem 2.1. Throughout this section, $\lambda \in (1/2, 1)$ is fixed, and we abbreviate $\mu = \mu_\lambda$, $X = X_\lambda$, and $s_u = s_u(\lambda)$. The first step is to turn concentration of μ into many close prefix sums with opposite first bits. Opposite first bits ensure that their difference polynomial has constant coefficient -1 , as required by the algebraic criterion.

3.1 Overlap of the first-bit measures

Define subprobability measures

$$\mu_i(A) = \mathbb{P}(X \in A, u_0 = i), \quad i = 0, 1.$$

Their sum is μ . The following fact uses only $\lambda > 1/2$, not singularity.

Lemma 3.1. *There is a finite measure $\xi \neq 0$ with $\xi \leq \mu_0$ and $\xi \leq \mu_1$.*

Proof. If no such measure exists, the densities of μ_0 and μ_1 with respect to μ have disjoint supports almost everywhere. Thus μ_0 and μ_1 are mutually singular. We will show that this makes the conditional entropy of the first n bits, given a λ^n -quantization of X , arbitrarily small compared with n . There are only $O(\lambda^{-n})$ quantization cells, too few to carry the full entropy $n \log 2$ of those bits. For every $\eta \in (0, 1/2)$, inner regularity then gives disjoint compact sets K_0, K_1 such that

$$\mathbb{P}(X \in K_{u_0}) > 1 - \eta.$$

Let $\delta = \text{dist}(K_0, K_1) > 0$, and choose an integer $h \geq 1$ with $\lambda^h < \delta/3$.

Set $Q_n = \lfloor X/\lambda^n \rfloor$. Given Q_n and the bits $u_0, \dots, u_{i-1}$, one can approximate

$$X^{(i)} = \frac{X - \sum_{k < i} u_k \lambda^k}{\lambda^i}$$

to within λ^{n-i} . The pair $(X^{(i)}, u_i)$ has the same law as (X, u_0) . For $i \leq n - h$, the approximation therefore determines an estimator of u_i with error probability at most η : when $X^{(i)} \in K_{u_i}$, choose the nearer of the two compact sets.

Write H for Shannon entropy using natural logarithms, and put $h_2(\eta) = -\eta \log \eta - (1-\eta) \log(1-\eta)$. The error-indicator argument gives

$$H(u_i \mid Q_n, u_0, \dots, u_{i-1}) \leq h_2(\eta) \quad (i \leq n - h).$$

Indeed, the estimator and its binary error indicator recover u_i ; the entropy of that indicator is at most $h_2(\eta)$. The remaining at most h bits have conditional entropy at most $h \log 2$. Since $X \in [0, L]$, the variable Q_n takes at most $L\lambda^{-n} + 2$ values. The chain rule now gives

$$n \log 2 \leq H(Q_n) + H(u_0, \dots, u_{n-1} \mid Q_n) \leq n \log(1/\lambda) + nh_2(\eta) + O_{\lambda, \eta}(1).$$

Choose η so small that $h_2(\eta) < \log(2\lambda)$ and let $n \rightarrow \infty$. This is a contradiction. $\square$

Proposition 3.2 (Many close prefixes with opposite first bits). *If μ_λ is singular, there are constants $a \in (0, L/2)$ and $c > 0$, depending only on λ , with the following property. For every $Q > 1$ and all sufficiently large n , at least $c2^n$ words $u \in \{0, 1\}^n$ with $u_0 = 0$ each have at least $Q(2\lambda)^n$ distinct neighbors $v \in \{0, 1\}^n$ such that*

$$v_0 = 1, \quad |s_u - s_v| \leq (L - a)\lambda^n. \quad (3.1)$$

The constants a, c are independent of Q and n .

Proof. Fix ξ from Lemma 3.1, and write $b = \xi(\mathbb{R}) > 0$. The endpoints $0, L$ have μ -measure zero, because each requires all the bits to have one prescribed value. Choose $a \in (0, L/2)$ so that $\mu([a, L - a]) > 1 - b/8$.

For each n , let $X^{(n)} = \sum_{k \geq 0} u_{n+k} \lambda^k$, and restrict the first-bit measures by defining

$$\mu'_{i,n}(A) = \mathbb{P}(X \in A, u_0 = i, X^{(n)} \in [a, L - a]).$$

The sum of the masses discarded from μ_0, μ_1 is less than $b/8$, since $X^{(n)}$ has law μ .

Fix $Q > 1$. By Lemma 2.3, choose a compact set K with $\text{Leb}(K) < \varepsilon$ and $\mu(K) > 1 - \varepsilon$, where

$$\varepsilon < b/8, \quad \frac{2\varepsilon Q}{a} < \frac{b}{4}.$$

Then $\xi(K) > 7b/8$. Partition the line into half-open intervals of length $a\lambda^n$, and let $\mathcal{J}_n$ consist of those meeting K . Compactness of K implies, for all sufficiently large n ,

$$\#\mathcal{J}_n \leq \frac{2\varepsilon}{a\lambda^n}.$$

To see this, their union is contained in the closed $a\lambda^n$ -neighborhood of K , whose Lebesgue measure tends to $\text{Leb}(K)$. Moreover,

$$\sum_{J \in \mathcal{J}_n} \min_{i=0,1} \mu'_{i,n}(J) \geq \frac{3b}{4}. \quad (3.2)$$

Indeed, the corresponding sum with μ_i in place of $\mu'_{i,n}$ is at least $\xi(K)$, and restricting the two measures loses less than $b/8$ in total.

Let $U_i(J)$ be the set of words u of length n with $u_0 = i$ for which

$$(s_u + \lambda^n[a, L - a]) \cap J \neq \emptyset.$$

Each prefix has probability 2^{-n} , so $\mu'_{i,n}(J) \leq 2^{-n} \#U_i(J)$. Discard the intervals for which $\#U_1(J) < Q(2\lambda)^n$. Their contribution to (3.2) is at most

$$\frac{2\varepsilon}{a\lambda^n} 2^{-n} Q(2\lambda)^n = \frac{2\varepsilon Q}{a} < \frac{b}{4}.$$

Thus the sum of the minima over the remaining intervals is at least $b/2$.

A fixed interval $s_u + \lambda^n[a, L - a]$ meets at most

$$C_a = \left\lceil \frac{L - 2a}{a} \right\rceil + 2$$

of the bins. If $\mathcal{U}$ is the union of $U_0(J)$ over the retained bins, it follows that

$$\frac{b}{2} \leq 2^{-n} \sum_{J \text{ retained}} \#U_0(J) \leq 2^{-n} C_a \#\mathcal{U}.$$

Take $c = b/(2C_a)$. For every $u \in \mathcal{U}$, choose one retained bin J with $u \in U_0(J)$. All $v \in U_1(J)$ satisfy (3.1): two points in J differ by less than $a\lambda^n$, and two tails in $\lambda^n[a, L - a]$ differ by at most $(L - 2a)\lambda^n$. There are at least $Q(2\lambda)^n$ such neighbors. $\square$

3.2 Completion to irreducible polynomials

For every pair in Proposition 3.2, the difference $w = u - v$ has $w_0 = -1$ and $|\sum_{i < n} w_i \lambda^i| \leq (L - a)\lambda^n$. We next extend any such coefficient vector to a polynomial whose root lies in $(\lambda, \lambda + 4^{-n}/2]$. The first lemma ensures that a bounded-coefficient polynomial cannot remain too flat throughout this short interval.

Lemma 3.3 (Uniform finite-order nonvanishing). *Fix $\lambda \in (0, 1)$. There are an integer $M \geq 1$ and constants $c_0, x_0 > 0$ such that every real polynomial $p(t) = -1 + \sum_{k \geq 1} a_k t^k$ with $|a_k| \leq 1$ satisfies*

$$\sup_{t \in [\lambda, \lambda + x]} |p(t)| \geq c_0 x^M \quad (0 < x < x_0).$$

Proof. There exist $M \geq 1$ and $\delta > 0$ such that

$$\max_{0 \leq k \leq M} \left| \frac{p^{(k)}(\lambda)}{k!} \right| \geq \delta \quad (3.3)$$

for every polynomial in the stated class. Otherwise, for each m one could choose a polynomial for which all these quantities through order m are less than $1/m$. Extend its coefficient sequence by zeros. A coefficientwise convergent subsequence exists by compactness of $[-1, 1]^{\mathbb{N}}$. The associated series converge, with every derivative, uniformly on compact subsets of the unit disk. Their limit would have all derivatives zero at λ and constant coefficient -1 , contradicting the identity theorem.

For fixed M , equivalence of norms on polynomials of degree at most M gives $\kappa_M > 0$ such that

$$\sup_{0 \leq t \leq 1} \left| \sum_{k=0}^M b_k t^k \right| \geq \kappa_M \max_{k \leq M} |b_k|.$$

Apply this to the Taylor polynomial of $p(\lambda + xt)$. For $x \leq 1$, (3.3) makes the right-hand side at least $\kappa_M \delta x^M$. The coefficient bound gives a uniform bound on the $(M+1)$ st derivative in a fixed neighborhood of λ , so the Taylor remainder is $O_\lambda(x^{M+1})$, with M fixed. Reducing x_0 proves the assertion with $c_0 = \kappa_M \delta / 2$. $\square$

To impose irreducibility without disturbing the prescribed coefficients, we use the prime-polynomial theorem in arithmetic progressions. We include the deduction needed here from the function-field Riemann hypothesis; its character-sum formulation is recalled in [6, Theorem 3.2 and Lemma 3.3].

Lemma 3.4 (Irreducibles with prescribed initial coefficients). *Fix integers $D \geq 2$ and $d > 2D$. For all sufficiently large n , every polynomial $F \in \mathbb{F}_2[t]$ with $\deg F < Dn$ and $F(0) = 1$ has a monic irreducible completion $P \in \mathbb{F}_2[t]$ satisfying*

$$\deg P = dn, \quad P \equiv F + t^{Dn} \pmod{t^{Dn+1}}.$$

The threshold in n is uniform in F .

Proof. For a monic polynomial f , write $\Lambda(f) = \deg P$ if $f = P^k$ for a monic irreducible P , and $\Lambda(f) = 0$ otherwise. Let χ be a nonprincipal Dirichlet character modulo t^ℓ , extended by zero to nonunits. Its L -function is a polynomial of degree at most $\ell - 1$, with inverse roots α_j satisfying $|\alpha_j| \leq \sqrt{2}$; this bound includes trivial factors. Taking the logarithmic derivative of the Euler product gives

$$\left| \sum_{\substack{f \text{ monic} \\ \deg f = s}} \Lambda(f) \chi(f) \right| = \left| \sum_j \alpha_j^s \right| \leq \ell 2^{s/2}.$$

For the principal character, the same sum is $2^s - 1$, by the zeta function of $\mathbb{F}_2[t]$ after removal of the prime t . Character orthogonality therefore gives, in every invertible residue class A modulo t^ℓ ,

$$\sum_{\substack{f \text{ monic, } \deg f = s \\ f \equiv A \pmod{t^\ell}}} \Lambda(f) \geq \frac{2^s - 1}{2^{\ell-1}} - \ell 2^{s/2}. \quad (3.4)$$

The contribution from proper prime powers, even without the congruence restriction, is at most

$$\sum_{\substack{k|s \\ k \geq 2}} 2^{s/k} \leq s 2^{s/2}.$$

Take $\ell = Dn + 1$, $s = dn$, and $A = F + t^{Dn}$. This is an invertible class. The main term in (3.4) has order $2^{(d-D)n}$, whereas the character error and proper-power contribution are $O(n2^{dn/2})$. Since $d > 2D$, at least one irreducible polynomial remains for all sufficiently large n , uniformly in A . $\square$

Proposition 3.5 (Realization of a polynomial prefix). *Fix $\lambda \in (1/2, 1)$ and $a \in (0, L)$, where $L = (1 - \lambda)^{-1}$. There is an integer d_0 such that, for each integer $d \geq d_0$ and all sufficiently large n , the following assertion holds uniformly over $w \in \{-1, 0, 1\}^n$. If*

$$w_0 = -1, \quad \left| \sum_{i=0}^{n-1} w_i \lambda^i \right| \leq (L - a) \lambda^n, \quad (3.5)$$

then some $r \in R(n, d)$ satisfies

$$\lambda < r \leq \lambda + \frac{1}{2} 4^{-n}, \quad \pi_n(p_r) = w.$$

Proof. We first append digits to make the polynomial small at λ . We then construct two irreducible completions with opposite signs there but the same sign at a nearby point to its right. Uniform finite-order nonvanishing supplies that second point, and the intermediate value theorem supplies a root between the two points.

Let M, c_0, x_0 be supplied by Lemma 3.3. Choose an integer $D \geq 2$ with

$$\lambda^D < 4^{-M}, \quad (3.6)$$

and then take $d_0 = 2D + 1$.

Start with the polynomial in (3.5) and append coefficients through index $Dn - 1$. If the next index is k , let e be the negative of the current value at λ , divided by λ^k . Appending $e_0 \in \{-1, 0, 1\}$ changes this residual to

$$e \mapsto \frac{e - e_0}{\lambda}.$$

Choose a nearest digit e_0 . Once $|e| \leq 1$, the next residual has absolute value at most $1/(2\lambda) < 1$, so this bound is preserved. Before that time, the residual keeps its sign and its absolute value updates by

$$|e| \mapsto \frac{|e| - 1}{\lambda} = L + \frac{|e| - L}{\lambda}.$$

The initial bound $|e| \leq L - a$ implies that $|e| \leq 1$ is reached within a number of steps depending only on λ, a . Indeed, after k steps without reaching this interval the absolute value is at most $L - a\lambda^{-k}$, which eventually falls below 1. Consequently, for all sufficiently large n , the polynomial p_0 constructed through index $Dn - 1$ satisfies

$$|p_0(\lambda)| \leq \frac{\lambda^{Dn}}{2\lambda}. \quad (3.7)$$

Put $x_n = 4^{-n}/2$. By Lemma 3.3, there is $t_n \in [\lambda, \lambda + x_n]$ with $|p_0(t_n)| \geq c_0 x_n^M$. Meanwhile every possible appendage with coefficients of absolute value at most one, starting at index Dn , is bounded throughout this interval by

$$\frac{(\lambda + x_n)^{Dn}}{1 - \lambda - x_n} = o(x_n^M).$$

This follows from (3.6), since $Dn x_n \rightarrow 0$. The estimate is uniform in the prefix and in the eventual length of the appendage.

Fix $d \geq d_0$. Apply Lemma 3.4 to the reduction of p_0 modulo 2, obtaining a monic irreducible polynomial P of degree dn with $P \equiv p_0 + t^{Dn} \pmod{t^{Dn+1}}$. Represent its coefficients of indices $Dn, \dots, dn$ by integers $b_k \in \{0, 1\}$, and let

$$T(t) = \sum_{k=Dn}^{dn} b_k t^k, \quad p_{\pm}(t) = p_0(t) \pm T(t).$$

Here $b_{Dn} = b_{dn} = 1$. Thus $T(\lambda) \geq \lambda^{Dn}$, and (3.7) implies $p_-(\lambda) < 0 < p_+(\lambda)$. At t_n , the negligible-tail estimate shows that both polynomials have the same nonzero sign as $p_0(t_n)$. One of them therefore has a real root

$$r \in (\lambda, t_n] \subset (\lambda, \lambda + x_n].$$

Both $p_{\pm}$ reduce to P modulo 2, so they are irreducible over $\mathbb{Q}$. They have degree dn , leading coefficient ± 1 , constant coefficient -1 , and all coefficients in $\{-1, 0, 1\}$. Their roots are algebraic units. For sufficiently large n , the chosen r is in $(1/2, 1)$, belongs to $R(n, d)$, and its signed minimal polynomial is the chosen $p_{\pm}$. Its first n coefficients are w . $\square$

3.3 Proof of the converse and the quantifiers

Suppose μ_{λ} is singular. Proposition 3.2 gives constants $a, c > 0$. Choose one integer d large enough for Proposition 3.5 and also satisfying $1/d < c$. This choice depends on λ alone.

Fix any integer $j \geq 1$ and choose $Q > 2j$. For all sufficiently large n , each of at least $c2^n$ words u has at least $Q(2\lambda)^n$ neighbors v supplied by Proposition 3.2. For each difference $w = u - v$, Proposition 3.5 supplies a root $r_w \in R(n, d)$ with signed minimal polynomial projecting to w , and

$$\lambda < r_w \leq \lambda + \frac{1}{2}4^{-n}.$$

Choose any one of these roots as a center r . All of them lie within 4^{-n} of r , so all their projected vectors belong to $W_n(r; d)$. For fixed u , its different neighbors give distinct vectors $u - v$, each in $\mathcal{O}_u$. Furthermore,

$$1 \leq \left(\frac{r}{\lambda}\right)^n \leq \left(1 + \frac{4^{-n}}{2\lambda}\right)^n \rightarrow 1$$

uniformly in the choice of center. Hence, for all sufficiently large n , each of these $c2^n$ orthants contains at least $j(2r)^n$ distinct vectors of $W_n(r; d)$. Since $c > 1/d$, we have $r \in R_*(n, d, j)$ and $\lambda \in [r - 4^{-n}, r)$.

We have proved that a single d works for every j , at every sufficiently large n . This proves membership in the expression (2.3) with $\liminf$. Membership in that expression implies membership with $\limsup$, which implies singularity by Section 2.2. Lemma 2.2 handles $\lambda \leq 1/2$ and supplies the absolute continuity of every remaining parameter. This completes the proof of Theorem 2.1.

4 Recovery of classical parameter classes

The arithmetic criterion is compatible with the classical norm separation argument for Garsia parameters and the Fourier argument for Pisot parameters. The first example directly bounds the crowding in (2.2); the second establishes singularity and then uses Theorem 2.1 to obtain the approximation property.

Proposition 4.1 (Garsia parameters, [4]). *Let $\beta \in (1, 2)$ be an algebraic integer of absolute norm 2 whose conjugates all have modulus greater than 1. Then $1/\beta \notin \mathcal{C}$, and $\nu_{1/\beta}$ is absolutely continuous.*

Proof. Put $\lambda = 1/\beta$. Consider distinct vectors $w, w' \in \{-1, 0, 1\}^n$ lying in one closed orthant. Their difference has coefficients in $\{-1, 0, 1\}$, so

$$P(t) = \sum_{i=0}^{n-1} (w_i - w'_i) t^{n-1-i}$$

is a nonzero polynomial of that coefficient type. It cannot vanish at β . Otherwise, after removing any power of t dividing P , its constant coefficient would be ± 1 , while divisibility by the monic minimal polynomial of β would force that coefficient to be divisible by 2.

For every other conjugate γ of β ,

$$|P(\gamma)| \leq \frac{|\gamma|^n}{|\gamma| - 1}.$$

The nonzero algebraic integer $P(\beta)$ has absolute norm at least one. Since the product of the moduli of all conjugates of β is 2, the preceding estimate gives a constant $c_\beta > 0$ such that

$$\left| \sum_{i=0}^{n-1} (w_i - w'_i) \lambda^i \right| = \beta^{-(n-1)} |P(\beta)| \geq c_\beta 2^{-n}. \quad (4.1)$$

For any potential hit of (2.3) at a sufficiently large n , all projected vectors satisfy the slab bound (2.5). Within any one orthant, (4.1) bounds their number by

$$1 + \frac{2B\lambda^n}{c_\beta 2^{-n}} \leq C_\beta (2\lambda)^n.$$

Choose an integer $j > C_\beta$. Since a hit has $r > \lambda$, no orthant can then meet the threshold $j(2r)^n$. This excludes all sufficiently large n , for every degree multiplier d . Thus $\lambda \notin \mathcal{C}$, and Theorem 2.1 gives absolute continuity. $\square$

Proposition 4.2 (Reciprocal Pisot parameters, [3]). *If $\beta \in (1, 2)$ is a Pisot number, then $1/\beta \in \mathcal{C}$, and $\nu_{1/\beta}$ is singular.*

Proof. A Pisot number is an algebraic integer greater than one whose other conjugates have modulus less than one. Here its nonzero integer norm has absolute value less than $\beta < 2$, so β is a unit. Consequently β^ℓ is an algebraic integer for every $\ell \in \mathbb{Z}$, and it can never be a half-integer of odd numerator. In particular, all factors $\cos(\pi\beta^\ell)$ are nonzero.

For negative ℓ , the quantity β^ℓ tends exponentially to zero. For positive ℓ , its distance to an integer tends exponentially to zero: the trace of β^ℓ is an integer and the contributions of the other conjugates decay exponentially. The quadratic behavior of $-\log |\cos(\pi t)|$ near the integers therefore gives

$$\sum_{\ell \in \mathbb{Z}} -\log |\cos(\pi\beta^\ell)| < \infty.$$

With $\lambda = 1/\beta$, independence of the signs yields

$$|\mathbb{E} \exp(i\pi\beta^n Y_\lambda)| = \prod_{\ell \leq n} |\cos(\pi\beta^\ell)| \geq \prod_{\ell \in \mathbb{Z}} |\cos(\pi\beta^\ell)| > 0.$$

The Riemann–Lebesgue lemma rules out absolute continuity. Lemma 2.2 gives singularity, and Theorem 2.1 gives $\lambda \in \mathcal{C}$. $\square$

5 Degree-four Salem parameters

For a degree-four Salem number β , the two conjugates on the unit circle form a single complex-conjugate pair. We use simultaneous control of this pair to construct many Fourier phases with appreciable expectation. After removing integer trace terms, widely separated phases depend, up to a small error, on disjoint blocks of the Bernoulli signs. Their averages therefore distinguish the Bernoulli convolution from Lebesgue measure.

Theorem 5.1. *Let $\beta \in (1, 2)$ be a Salem number of degree four. Then the unbiased Bernoulli convolution $\nu_{1/\beta}$ is singular with respect to Lebesgue measure.*

The proof has two quantitative requirements. The Fourier products must lose little mass before the algebraic trace becomes an accurate integer approximation; the remaining trace errors must then stay small over many disjoint blocks. The multiplier construction below supplies both requirements. The trace approximation is the familiar arithmetic ingredient in the Fourier analysis of Pisot and Salem parameters [3]; see in particular Salem's small-conjugate construction [15, Chapter III, Section 3, Theorem III, pp. 28–29]. The estimates here also control the losses accumulated while the multiplier changes.

5.1 Multipliers with small unit-circle conjugates

Fix β as in Theorem 5.1, and write its conjugates as $\beta^{-1}, z, \bar{z}$, where $z = e^{i\theta}$. The number β is a unit: the product of its four conjugates is 1. Moreover, θ/π is irrational, since otherwise z , and hence its conjugate β , would be a root of unity.

For positive integers $k_1, k_2, \dots$, to be chosen, define

$$\begin{aligned} A_m &= \prod_{h=1}^m (\beta^{k_h} - \beta^{-k_h}), & b_m &= \sum_{h=1}^m k_h, \\ c_m &= \prod_{h=1}^m (z^{k_h} - z^{-k_h}), & \delta_m &= |c_m|, & L_m &= \log(1/\delta_m). \end{aligned} \tag{5.1}$$

Here and throughout this section logarithms are natural. Irrationality ensures $\delta_m > 0$, and $0 < A_m \leq \beta^{b_m}$. For $j \in \mathbb{Z}$, put $u_{m,j} = A_m \beta^j$. All these numbers are algebraic integers, because β is a unit. Their field traces are the integers

$$T_{m,j} = u_{m,j} + (-1)^m u_{m,-j} + 2 \operatorname{Re}(c_m z^j) \in \mathbb{Z}. \tag{5.2}$$

Indeed, the embedding taking β to β^{-1} takes A_m to $(-1)^m A_m$. Also $u_{m,j} \notin \mathbb{Z} + \frac{1}{2}$, since a rational algebraic integer is an integer. Thus none of the cosine factors used below vanishes.

The construction below balances the decrease of δ_m against the cosine losses generated by A_m . Eventually L_m grows by a factor greater than two while $k_{m+1} \delta_m^2$ tends to zero; these are the margins used in the loss estimate below. It also keeps b_m small compared with δ_m^{-1} , leaving room for many separated blocks of signs. A uniform lower bound on the cosines will permit a quadratic estimate for perturbations of their logarithms.

Lemma 5.2. *There are positive integers k_m and a constant $\gamma > 0$ for which the numbers L_m in (5.1) are positive and satisfy the following properties. Set*

$$D = \frac{10}{\log \beta}, \quad K_m = \lceil 2b_m + DL_m \rceil \quad (m \geq 2).$$

Then $A_m \geq 1$ for every m , and

$$K_m \leq k_{m+1} \leq e^{1.9L_m}, \quad L_{m+1} \geq 1.75L_m \quad (m \geq 2), \quad (5.3)$$

$$\log K_m \leq .75L_m, \quad L_{m+1} \geq 2.1L_m \quad (m \geq 3). \quad (5.4)$$

Moreover,

$$|\cos(\pi u_{m,j})| \geq \gamma \quad (m \geq 1, j \in \mathbb{Z}), \quad (5.5)$$

and, for $m \geq 2$,

$$2\delta_m \leq \frac{\gamma}{2\pi}, \quad \beta^{b_m - k_{m+1}} \leq e^{-10L_m} \leq \frac{\gamma}{2\pi}. \quad (5.6)$$

Proof. Write $\|x\|_{\mathbb{R}/\mathbb{Z}}$ for distance to the nearest integer. The pigeonhole principle gives, for every integer $N \geq 1$, an integer $q \in [1, N]$ such that

$$\|q\theta/\pi\|_{\mathbb{R}/\mathbb{Z}} \leq N^{-1}. \quad (5.7)$$

For example, place the $N + 1$ fractional parts of $0, \theta/\pi, \dots, N\theta/\pi$ in N intervals of length $1/N$. Since θ/π is irrational, the possible q in (5.7) are unbounded as $N \rightarrow \infty$. In particular there are arbitrarily large q with $\|q\theta/\pi\|_{\mathbb{R}/\mathbb{Z}} \leq 1/q$.

Choose and fix k_1 so that $A_1 > 1$ and $\delta_1 < 1/16$. Define

$$\gamma_m = \inf_{j \in \mathbb{Z}} |\cos(\pi u_{m,j})|.$$

At this point $\gamma_1 > 0$: as $j \rightarrow -\infty$, $u_{1,j} \rightarrow 0$; as $j \rightarrow +\infty$, the trace identity shows that the absolute cosine differs from $|\cos(\pi u_{1,-j})|$ by at most $2\pi\delta_1 < \pi/8$. The remaining finite set of factors contains no zero.

Next choose $k_2 > k_1$ from the unbounded sequence satisfying $\|k_2\theta/\pi\|_{\mathbb{R}/\mathbb{Z}} \leq 1/k_2$. We will impose finitely many lower thresholds on this choice. Since

$$\delta_2 \leq \frac{2\pi\delta_1}{k_2}, \quad L_2 \geq \log k_2 - \log(2\pi\delta_1),$$

both k_2 and L_2 can be made arbitrarily large. In particular, $\log K_2 \leq 1.1L_2$ for every sufficiently large such choice: the last inequality bounds k_2 by a fixed multiple of e^{L_2} , whereas the other contribution DL_2 to K_2 is only linear in L_2 .

For $m \geq 2$, suppose $L = L_m$ is large and $\log K_m \leq 1.1L$. Take $N = \lfloor e^{1.9L} \rfloor$ and choose q by (5.7). Define

$$k_{m+1} = \begin{cases} q, & q \geq K_m, \\ \lceil K_m/q \rceil q, & q < K_m. \end{cases} \quad (5.8)$$

Let $L' = L_{m+1}$. If $q \geq K_m$, then

$$\delta_{m+1} \leq 2\pi\delta_m/N, \quad L' \geq 2.9L - O(1) \geq 2.8L, \quad b_{m+1} \leq 2e^{1.9L}.$$

If $q < K_m$, the multiplier $\lceil K_m/q \rceil$ is at most $2K_m$ and $K_m \leq k_{m+1} < 2K_m$. Using $|\sin(\pi x)| \leq \pi\|x\|_{\mathbb{R}/\mathbb{Z}}$ gives

$$\delta_{m+1} \leq 4\pi\delta_m K_m/N, \quad L' \geq 2.9L - \log K_m - O(1) \geq 1.75L,$$

and $b_{m+1} \leq 3e^{1.1L}$. In both cases $K_m \leq k_{m+1} \leq e^{1.9L}$ when L is sufficiently large.

These estimates also give $\log K_{m+1} \leq .75L'$. To see this without an upper bound on L' , write

$$K_{m+1} \leq 2b_{m+1} + DL' + 1.$$

The bound on b_{m+1} makes the first term at most $\frac{1}{2}e^{.75L'}$ for large L , since $1.9 < .75 \cdot 2.8$ in the first case and $1.1 < .75 \cdot 1.75$ in the second. The remaining term satisfies the same bound for every sufficiently large L' . Thus the construction iterates. From $m = 3$ onward, the improved bound $\log K_m \leq .75L_m$ gives $L_{m+1} \geq (2.9 - .75)L_m - O(1) \geq 2.1L_m$ in the second case; the first case already gives more. This proves (5.3)–(5.4) after a lower threshold on L_2 depending only on β .

It remains to choose k_2 large enough to obtain the cosine floor. For $j \leq 0$ and $k = k_{m+1}$, the exact identity

$$u_{m+1,j} = u_{m,j+k} - u_{m,j-k}, \quad |u_{m,j-k}| \leq \beta^{b_m-k} \beta^j \quad (5.9)$$

and the Lipschitz bound for the cosine give a lower bound $\gamma_m - \pi\beta^{b_m-k}$. For positive j , reflect to $-j$ using (5.2) at level $m+1$. Consequently

$$\gamma_{m+1} \geq \gamma_m - \pi\beta^{b_m-k_{m+1}} - 2\pi\delta_{m+1}. \quad (5.10)$$

For $m \geq 2$, the inequality $k_{m+1} \geq 2b_m + DL_m$ implies $\beta^{b_m-k_{m+1}} \leq e^{-10L_m}$. Also $L_m \geq 1.75^{m-2}L_2$. Thus the sum of the decrements in (5.10) is at most

$$\pi\beta^{k_1-k_2} + 2\pi e^{-L_2} + \sum_{r=0}^{\infty} \left(\pi e^{-10 \cdot 1.75^r L_2} + 2\pi e^{-1.75^{r+1} L_2} \right).$$

This tends to zero as $k_2 \rightarrow \infty$ along the chosen sequence. Choose k_2 so that this sum is less than $\gamma_1/10$, and so that $2e^{-L_2}$ and e^{-10L_2} are at most $\gamma_1/(4\pi)$. These requirements are compatible with the earlier lower thresholds. Now fix k_2 and the resulting sequence, and put $\gamma = \gamma_1/2$. Equations (5.5)–(5.6) follow. Finally, k_m is increasing and every factor $\beta^{k_m} - \beta^{-k_m}$ is at least its first factor $A_1 > 1$, so $A_m \geq 1$. $\square$

5.2 The loss in the cosine products

Fix the sequence furnished by Lemma 5.2. Constants denoted by C below may change from line to line, but are independent of m and of the phase index. Define

$$f(u) = -\log |\cos(\pi u)| \quad \text{for } u \notin \mathbb{Z} + \frac{1}{2}, \quad S_m = \sum_{j \leq 0} f(u_{m,j}).$$

Each S_m is finite: its terms have a geometrically decaying tail because $u_{m,j} \rightarrow 0$ as $j \rightarrow -\infty$ and $f(u) = O(u^2)$ near zero. The next estimate controls the entire negative-power contribution even though A_m becomes large.

Lemma 5.3. *For the sequence in Lemma 5.2, $S_m = o(L_m)$. Furthermore, with $\eta = .01$, for $m \geq 2$ and $n \geq 0$,*

$$\sum_{j \leq n} f(u_{m,j}) \leq (2 + \eta)S_m + Cn\delta_m^2. \quad (5.11)$$

Proof. We first record a perturbation estimate. If $|\cos(\pi u)| \geq \gamma$ and $|v| \leq \gamma/(2\pi)$, then

$$f(u+v) \leq (1 + \eta)f(u) + Cv^2. \quad (5.12)$$

Indeed, the absolute cosine is at least $\gamma/2$ along the segment from u to $u + v$, so f'' is bounded there. Also

$$|f'(u)| \leq \frac{\pi}{\gamma} \sqrt{2f(u)},$$

because $1 - |\cos(\pi u)|^2 \leq 2f(u)$. Taylor's formula and $|f'(u)v| \leq \eta f(u) + Cv^2$ prove (5.12).

The function f is even and 1-periodic. Reflecting positive indices by (5.2), and using (5.6), gives

$$f(u_{m,i}) \leq (1 + \eta)f(u_{m,-i}) + C\delta_m^2 \quad (m \geq 2, i \geq 1). \quad (5.13)$$

Summing this inequality proves (5.11).

To estimate S_{m+1} , apply (5.12) to (5.9), with base point $u_{m,j+k}$ and $v = -u_{m,j-k}$, where $k = k_{m+1}$. The cosine floor holds at every base point, including those with positive index. The squared perturbations have a geometric sum, so

$$\begin{aligned} S_{m+1} &\leq (1 + \eta) \left(S_m + \sum_{i=1}^k f(u_{m,i}) \right) + C\beta^{2(b_m-k)} \\ &\leq (1 + \eta)(2 + \eta)S_m + C(1 + k\delta_m^2). \end{aligned}$$

By (5.3), $k\delta_m^2 \leq e^{-.1L_m} \leq 1$. Therefore $S_{m+1} \leq rS_m + C$, where $r = (1.01)(2.01) = 2.0301$. The already fixed value S_2 is finite, so iteration gives $S_m = O(r^{m-2})$. In contrast, (5.4) gives $L_m \geq 2.1^{m-3}L_3$ for $m \geq 3$. Since $r < 2.1$, their ratio tends to zero. $\square$

5.3 Independent blocks of Fourier phases

We now turn the product estimates into sets of almost full $\nu_{1/\beta}$ -measure and vanishing Lebesgue measure. The comparison of phase averages under two measures is analogous to the separation argument for Riesz products in [13, Section 1.1, Theorem 1.2]. Here the needed estimates follow from the independent signs, as we now show. Work on the probability space of independent uniform signs $(\epsilon_\ell)_{\ell \geq 0}$, and write

$$Y = \sum_{\ell \geq 0} \epsilon_\ell \beta^{-\ell}, \quad W_{m,n} = \exp(i\pi A_m \beta^n Y) \quad (n \geq 0).$$

The law of Y is $\nu_{1/\beta}$, supported on $I = [-(1 - \beta^{-1})^{-1}, (1 - \beta^{-1})^{-1}]$. Independence, followed by passage to the limit in the convergent sign series, gives

$$\mathbb{E}W_{m,n} = \prod_{j=-\infty}^n \cos(\pi u_{m,j}). \quad (5.14)$$

This is a nonzero real number; the negative-index tail has a convergent sum of logarithmic losses.

Set

$$\begin{aligned} J_m &= \lfloor e^{1.5L_m} \rfloor, \quad H_m = \lceil b_m + DL_m \rceil, \\ \mathcal{N}_m &= \{H_m + s(2H_m + 1) : s \in \mathbb{Z}_{\geq 0}\} \cap [0, J_m], \end{aligned}$$

and let $M_m = |\mathcal{N}_m|$. All subsequent estimates concern sufficiently large m . Since $H_m \leq K_m \leq e^{.75L_m}$,

$$e^{.7L_m} \leq M_m \leq J_m + 1. \quad (5.15)$$

Lemma 5.3 and $J_m\delta_m^2 \leq e^{-.5L_m}$ show that

$$|\mathbb{E}W_{m,n}| \geq e^{-L_m/20} \quad (0 \leq n \leq J_m). \quad (5.16)$$

For $n \in \mathcal{N}_m$, retain only the signs whose indices lie in $[n - H_m, n + H_m]$, and define

$$\widetilde{W}_{m,n} = (-1)^{\sum_{j=H_m+1}^n T_{m,j}} \exp \left(i\pi \sum_{\ell=n-H_m}^{n+H_m} u_{m,n-\ell} \epsilon_\ell \right). \quad (5.17)$$

The integer sum in the prefactor is empty when $n = H_m$. To explain this approximation, expand the exponent of $W_{m,n}$ as $\pi \sum_{j \leq n} u_{m,j} \epsilon_{n-j}$. For $j > H_m$, replace $u_{m,j}$ by its integer trace $T_{m,j}$; the resulting factor is deterministic because $e^{i\pi T_{m,j} \epsilon_{n-j}} = (-1)^{T_{m,j}}$. For $j < -H_m$, drop the term. The remaining indices are exactly those in (5.17).

The difference of the two phase angles, after these replacements, is a sum of independent mean-zero signs. The inequality $|e^{ix} - e^{iy}| \leq |x - y|$ therefore gives

$$\begin{aligned} \|W_{m,n} - \widetilde{W}_{m,n}\|_2^2 &\leq \pi^2 \left(\sum_{j=H_m+1}^n |u_{m,j} - T_{m,j}|^2 + \sum_{j < -H_m} |u_{m,j}|^2 \right) \\ &\leq C \left(\beta^{2(b_m - H_m)} + n\delta_m^2 \right) \leq C e^{-L_m/2}. \end{aligned} \quad (5.18)$$

Here (5.2) bounds the first summand by a geometric tail and a contribution $Cn\delta_m^2$; the other sum is itself a geometric tail. Finally, $D \log \beta = 10$ and $n \leq J_m$ give the last inequality. All norms in (5.18) refer to the sign probability space. The windows belonging to consecutive members of $\mathcal{N}_m$ are disjoint, so the variables $\widetilde{W}_{m,n}$ are independent.

Proof of Theorem 5.1. For $n \in \mathcal{N}_m$, let $\sigma_{m,n} \in \{-1, 1\}$ be the sign of the real expectation in (5.14). Define the continuous function

$$F_m(y) = \frac{1}{M_m} \sum_{n \in \mathcal{N}_m} \sigma_{m,n} e^{i\pi A_m \beta^n y}.$$

Its mean at Y is real and, by (5.16), $a_m := \mathbb{E} F_m(Y) \geq e^{-L_m/20}$. The corresponding average of the independent variables $\sigma_{m,n} \widetilde{W}_{m,n}$ has centered L^2 norm at most $M_m^{-1/2}$, since each variable has modulus one. By the triangle inequality and (5.18),

$$\|F_m(Y) - a_m\|_2 \leq M_m^{-1/2} + C e^{-L_m/4}. \quad (5.19)$$

Subtracting the means of the approximation errors introduces at most a factor two, included in C .

On the other hand, the same average is small in Lebesgue $L^2(I)$. Write its frequencies in increasing order as $\xi_1 < \dots < \xi_{M_m}$. Since $A_m \geq 1$ and their exponents are distinct nonnegative integers, $\xi_{r+1} - \xi_r \geq \pi(\beta - 1)$. Thus $|\xi_r - \xi_s| \geq \pi(\beta - 1)|r - s|$ and

$$\left| \int_I e^{i(\xi_r - \xi_s)y} dy \right| \leq \frac{2}{|\xi_r - \xi_s|} \quad (r \neq s).$$

Expanding the square and summing by the rank difference yields

$$\int_I |F_m(y)|^2 dy \leq C \frac{1 + \log M_m}{M_m}. \quad (5.20)$$

Consider the compact sets

$$E_m = \{y \in I : \operatorname{Re} F_m(y) \geq \frac{1}{2} e^{-L_m/20}\}.$$

Chebyshev's inequality, (5.19), and (5.15) imply

$$1 - \nu_{1/\beta}(E_m) \leq 4e^{L_m/10} \left(M_m^{-1/2} + Ce^{-L_m/4} \right)^2 \longrightarrow 0.$$

Likewise, writing $|E_m|$ for Lebesgue measure,

$$|E_m| \leq 4e^{L_m/10} \int_I |F_m(y)|^2 dy \leq Ce^{L_m/10} \frac{1 + \log M_m}{M_m} \longrightarrow 0,$$

because $M_m \geq e^{7L_m}$ and $\log M_m \leq 1.5L_m + O(1)$. Choose a subsequence (m_r) along which $\sum_r (|E_{m_r}| + 1 - \nu_{1/\beta}(E_{m_r})) < \infty$. Then $E = \limsup_r E_{m_r}$ is Borel and Lebesgue null, since $|\bigcup_{r \geq R} E_{m_r}| \leq \sum_{r \geq R} |E_{m_r}| \rightarrow 0$. Moreover, the summability of the complementary probabilities shows that $\nu_{1/\beta}$ -almost every point lies in all sufficiently late E_{m_r} , and hence in E . Therefore $\nu_{1/\beta}(E) = 1$. $\square$

5.4 Two explicit parameters

Corollary 5.4. *Let β_1 and β_2 be the respective real roots greater than one of*

$$P_1(x) = x^4 - x^3 - x^2 - x + 1, \quad P_2(x) = x^4 - 2x^3 + x^2 - 2x + 1.$$

Then $\beta_1, \beta_2 \in (1, 2)$ and both ν_{1/β_1} and ν_{1/β_2} are singular.

Proof. For $y = x + x^{-1}$, the equations $P_i(x)/x^2 = 0$ become, respectively,

$$y^2 - y - 3 = 0, \quad y^2 - 2y - 1 = 0.$$

Their larger roots are $y_1 = (1 + \sqrt{13})/2$ and $y_2 = 1 + \sqrt{2}$, both in $(2, 5/2)$. For each i , the equation $x + x^{-1} = y_i$ has one root $\beta_i \in (1, 2)$ and its reciprocal. The other quadratic root y'_i lies in $(-2, 2)$, so the remaining two roots of P_i form a nonreal conjugate pair on the unit circle.

It remains to check that these four roots are algebraic conjugates. The quadratic field $\mathbb{Q}(y_i)$ is real, and the discriminant $y_i^2 - 4$ has negative image $(y'_i)^2 - 4$ under its nontrivial embedding. It cannot be a square in that field, since the image of a square under a real embedding is nonnegative. Hence $x^2 - y_i x + 1$ is irreducible over $\mathbb{Q}(y_i)$, and $[\mathbb{Q}(\beta_i) : \mathbb{Q}] = 4$. Thus β_1 and β_2 are degree-four Salem numbers, and Theorem 5.1 applies. $\square$

6 A polynomial with a weakly expanding conjugate pair

The final example uses a polynomial with one conjugate pair just outside the unit circle. We will prove singularity directly by counting typical prefixes. Counting all polynomial values in their conjugate coordinates therefore gives slightly too many possible prefixes. We obtain the required improvement by a product of positive trigonometric factors: its integral in the expanding coordinates stays bounded, whereas it grows exponentially on a set of words whose probability tends to one.

Throughout this section, every terminating decimal denotes the corresponding exact rational number. Put

$$p(t) = t^{31} - 2t^{30} + \sum_{j=0}^9 (t^{3j+1} - t^{3j}). \quad (6.1)$$

The identity

$$(t^2 + t + 1)p(t) = t^{33} - t^{32} - t^{31} - t^{30} - 1 \quad (6.2)$$

will control the near-cancellations in the trigonometric product.

Theorem 6.1. *The polynomial p in (6.1) has a unique real root $\beta \in (1.8392, 1.8393)$. This root is not a Pisot number, and $\nu_{1/\beta}$ is singular with respect to Lebesgue measure.*

We first record the root bounds needed in the proof. The proposition is proved by rational root disks in Section A.1.

Proposition 6.2 (Root bounds). *All roots of p are simple. Its roots outside the unit circle are $\beta, \alpha, \bar{\alpha}$, where β is real, $\text{Im } \alpha > 0$, and*

$$\begin{aligned} 1.8392 < \beta < 1.8393, \quad 1.0002089 < |\alpha|^2 < 1.0002094, \\ |\alpha|^{-1} < 0.9999. \end{aligned} \tag{6.3}$$

The remaining roots are fourteen nonreal conjugate pairs, each of modulus less than 0.994.

Write $D = \{\beta, \alpha, \bar{\alpha}\}$ and let I be the set of remaining roots. The expanding volume factor is

$$\mathcal{M} = \prod_{r \in D} |r| = \beta |\alpha|^2, \quad \log \mathcal{M} < \log \beta + 0.0002094. \tag{6.4}$$

The last inequality uses $\log(1+u) < u$ for $u > 0$.

The root bounds already show that β is not Pisot. Indeed, suppose β were Pisot, and divide p by its monic minimal polynomial. The quotient is monic and integral, has constant term of absolute value one, and contains $\alpha, \bar{\alpha}$ among its roots but not β . The product of the moduli of its roots is greater than one if it has no roots in I , and otherwise is at most $1.0002094 \cdot 0.994 < 1$. Both conclusions contradict the constant term. It remains to prove singularity.

6.1 Word vectors and characters

For a word $a = (a_0, \dots, a_{N-1}) \in \{0, 1\}^N$, define

$$y_r(a) = \sum_{j=0}^{N-1} a_j r^j \quad (r \in D \cup I), \quad x_r(a) = r^{-N} y_r(a) \quad (r \in D). \tag{6.5}$$

Vectors indexed by roots are always constrained by $y_{\bar{r}} = \overline{y_r}$, and a conjugate pair counts as one complex coordinate, or two real coordinates. In particular, the space of all y -vectors has real dimension 31.

Lemma 6.3. *All vectors $y(a)$, for all N and all words a , belong to a fixed full lattice in this real vector space. Their contracting coordinates $(y_r)_{r \in I}$ and normalized expanding coordinates $(x_r)_{r \in D}$ lie in fixed bounded boxes. The change from (x_D, y_I) to (y_D, y_I) has real Jacobian determinant of absolute value $\mathcal{M}^N$.*

Proof. Reduction of $\sum a_j t^j$ modulo the monic polynomial p gives an integer coefficient vector of length 31. Evaluation at the roots is an invertible real-linear map: a polynomial of degree at most 30 that vanishes at all 31 distinct roots is zero. The image of $\mathbb{Z}^{31}$ under this map is the required lattice. No irreducibility assumption is needed. The bounds follow from

$$|y_r(a)| \leq \frac{1}{1 - |r|} \quad (r \in I), \quad |x_r(a)| \leq \frac{1}{|r| - 1} \quad (r \in D).$$

Multiplication of the real coordinate by β^N and of the complex coordinate by α^N has determinant $\beta^N |\alpha|^{2N} = \mathcal{M}^N$. $\square$

To distinguish the word vectors from typical points of these boxes, define the real sequences

$$C_0(m) = \begin{cases} \sum_{r \in I} \frac{r^{-1-m}}{p'(r)}, & m < 0, \\ -\sum_{r \in D} \frac{r^{-1-m}}{p'(r)}, & m \geq 0, \end{cases} \quad C_1(m) = C_0(m+3) - C_0(m) + C_0(m-1). \quad (6.6)$$

Partial fractions give

$$\frac{1}{p(t)} = \sum_{m \in \mathbb{Z}} C_0(m) t^m, \quad \max_{r \in I} |r| < |t| < \min_{r \in D} |r|.$$

Thus C_0 is the bilateral Laurent coefficient sequence of $1/p$, and both C_0, C_1 decay exponentially at both ends. The same inversion by an exponentially decaying two-sided sequence is used by Lind and Schmidt [10, Lemma 4.5 and Example 4.7] to construct homoclinic points for algebraic actions. Here the explicit root formula will give characters adapted to the word vectors.

For arbitrary expanding coordinates x_D , with $x_\beta \in \mathbb{R}$ and $x_{\bar{\alpha}} = \overline{x_\alpha}$, put

$$\begin{aligned} Z_i^{(0,N)}(x) &= \exp\left(-2\pi i \sum_{r \in D} \frac{x_r r^i}{p'(r)}\right) \exp\left(-\pi i \sum_{k=0}^{N-1} C_0(k-i)\right), \\ Z_i^{(1,N)}(x) &= Z_{i-3}^{(0,N)}(x) \overline{Z_i^{(0,N)}(x)} Z_{i+1}^{(0,N)}(x), \quad i \in \mathbb{Z}. \end{aligned} \quad (6.7)$$

These functions have modulus one.

Lemma 6.4 (Characters at word vectors). *If $b_k = a_{N-1-k}$, then for $A \in \{0, 1\}$ and $i \in \mathbb{Z}$,*

$$Z_i^{(A,N)}(x(a)) = \exp\left(\pi i \sum_{k=0}^{N-1} C_A(k-i)(2b_k - 1)\right). \quad (6.8)$$

Proof. The reversed word gives $x_r(a) = \sum_{k=0}^{N-1} b_k r^{-1-k}$. For each integer $l \geq 0$, Lagrange interpolation shows that

$$\sum_{r \in D \cup I} \frac{r^l}{p'(r)}$$

is the coefficient of t^{30} in the remainder of t^l modulo p , and hence is an integer. Consequently, when $m < 0$, the coefficient $-\sum_{r \in D} r^{-1-m}/p'(r)$ differs from $C_0(m)$ by an integer. Its contribution to the exponent with coefficient $2\pi i b_k$ is unchanged by this replacement. When $m \geq 0$, the coefficients already agree. The second factor in (6.7) supplies the centering by -1 , proving the formula for $A = 0$. Multiplication of the three characters proves it for $A = 1$. $\square$

The corresponding stationary model uses independent fair bits $(b_k)_{k \in \mathbb{Z}}$ and the unit random variables

$$\mathcal{Z}_i^{(A)} = \exp\left(\pi i \sum_{k \in \mathbb{Z}} C_A(k-i)(2b_k - 1)\right). \quad (6.9)$$

The series converges absolutely. Independence gives the real moments

$$\begin{aligned} P_A &= \mathbb{E} \mathcal{Z}_i^{(A)} = \prod_{m \in \mathbb{Z}} \cos(\pi C_A(m)), \\ P_{AB}^\pm(k) &= \mathbb{E}[\mathcal{Z}_i^{(A)} (\mathcal{Z}_{i-k}^{(B)})^{\pm 1}] \\ &= \prod_{m \in \mathbb{Z}} \cos(\pi(C_A(m) \pm C_B(m+k))), \quad k \geq 0. \end{aligned} \quad (6.10)$$

6.2 A predictor with a positive mean gain

The weakly expanding pair produces long decaying tails in C_0 . Some shifted tails nearly cancel in the signed products $P_{00}^\pm(k)$. A linear combination of the corresponding phases will use these correlations. For rational weights w_{ks} , define

$$\begin{aligned} G_i^{(N)}(x) &= -0.001 + 0.0055 Z_{i-33}^{(1,N)}(x) + \sum_{k=30}^{2000} \sum_{s=0}^1 w_{ks} (Z_{i-k}^{(0,N)}(x))^{1-2s}, \\ \mathcal{G}_i &= -0.001 + 0.0055 \mathcal{Z}_{i-33}^{(1)} + \sum_{k=30}^{2000} \sum_{s=0}^1 w_{ks} (\mathcal{Z}_{i-k}^{(0)})^{1-2s}, \\ \mathcal{L}_i &= \operatorname{Re}(\mathcal{Z}_i^{(0)} \mathcal{G}_i). \end{aligned} \tag{6.11}$$

Proposition 6.5 (Predictor estimates). *There exist weights $w_{ks} \in 10^{-9}\mathbb{Z}$, indexed by $30 \leq k \leq 2000$ and $s \in \{0, 1\}$, such that*

$$0.001 + 0.0055 + \sum_{k,s} |w_{ks}| = \frac{198442946}{10^9} < x_* := 0.199 \tag{6.12}$$

and, in the stationary model,

$$\begin{aligned} \mathbb{E}\mathcal{L}_i &> 0.000210, \quad \mathbb{E}|\mathcal{G}_i|^2 < 0.000145, \\ \left| \mathbb{E}[(\mathcal{Z}_i^{(0)})^2 \mathcal{G}_i^2] \right| &< 0.000007. \end{aligned} \tag{6.13}$$

In particular, $|G_i^{(N)}(x)| \leq x_*$ for every N, i, x , and $|\mathcal{G}_i| \leq x_*$ for every bit sequence.

We fix weights supplied by Proposition 6.5. Their exact rational construction is given in Section A.2, and the moment bounds are proved in Section A.4.

The next two subsections construct a positive product whose logarithm has a positive mean under this stationary model and whose integral is bounded in the expanding coordinates. These two estimates will remove an exponential proportion of the lattice points allowed by Lemma 6.3.

6.3 Damping the possible frequency cancellations

Choose a fixed integer $i_0 \geq 1$ so that

$$\frac{0.1\beta^{i_0}}{|p'(\beta)|} > 1. \tag{6.14}$$

For a word length N , use every complete block

$$B_q = \{i_0 + 22q, \dots, i_0 + 22q + 20\} \subseteq \{0, \dots, N-1\}, \quad q \geq 0,$$

and write J_N for their union. Thus each block has 21 indices and consecutive blocks are separated by one unused index.

The near-relation (6.2) dictates which signed digit patterns need to be retained when integrating a block. For a pattern $(d_i)_{i \in B_q} \in \{-1, 0, 1\}^{B_q}$, let $b = \max B_q$ and set

$$v_i = \beta^{-i} \sum_{j=i}^b d_j \beta^j, \quad v_{b+1} = 0.$$

Reading from the largest index downwards gives $v_i = \beta v_{i+1} + d_i$. At the root β , (6.2) becomes

$$\beta^3 - \beta^2 - \beta - 1 = \beta^{-30}. \quad (6.15)$$

In particular, the four digits $[\sigma, -\sigma, -\sigma, -\sigma]$ leave the small normalized remainder $\sigma\beta^{-30}$. We encode these approximate returns by the transitions in Figure 1. The initial state is 0, and the permitted terminal states are $0, C_+, C_-$. For each sign $\sigma \in \{-1, 1\}$, the numerical values assigned to the states are

$$0, \quad A_\sigma : \sigma, \quad B_\sigma : \sigma(\beta - 1), \quad C_\sigma : \sigma(\beta^2 - \beta - 1). \quad (6.16)$$

A pattern is called *admissible* if every transition is allowed and its terminal state is permitted.

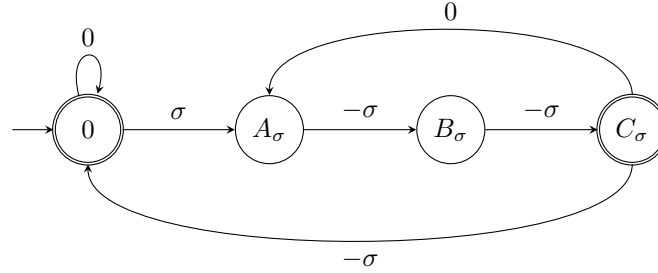


Figure 1: Admissible digits, read from high exponents to low exponents. There are two copies of the three nonzero states, one for each sign σ , sharing state 0. Edge labels are digits; double circles mark permitted terminal states. A return to 0 may be followed by either sign.

Let $o(d) = \#\{i : d_i \neq 0\}$. At a position i of a block, define

$$\eta_i = \sum_{\substack{d \text{ admissible} \\ d_i \neq 0}} \frac{x_*^{o(d)-2}}{o(d)}. \quad (6.17)$$

The same coefficients are repeated in every block. An admissible nonzero pattern has at least three nonzero digits, so all summands are well defined.

Lemma 6.6 (Cost of admissible patterns). *Set*

$$S_* = \left(1 + \frac{2x_*^4}{1 - x_*^2}\right)^{18} \left(1 + \frac{2x_*^3}{1 - x_*^2}\right) - 1. \quad (6.18)$$

Then $S_* < 0.081$, and

$$\begin{aligned} \sum_{\substack{d \text{ admissible} \\ o(d) > 0}} x_*^{o(d)} &\leq S_*, & \eta_i x_*^2 &\leq S_*/3, \\ \frac{\bar{\eta}}{1 - S_*/3} &< 0.101, & \bar{\eta} &= \frac{1}{21} \sum_{i \in B_q} \eta_i. \end{aligned} \quad (6.19)$$

Proof. A completed excursion from 0 to 0 has the digit string

$$[\sigma, -\sigma, -\sigma, (0, -\sigma, -\sigma)^l, -\sigma], \quad l \geq 0.$$

It has $4 + 2l$ nonzero digits and can begin in at most 18 positions of a block. An optional final excursion ending at C_σ has the same form without the last digit; its sign and length determine

its starting position. Ignoring incompatibilities between these choices overcounts the admissible patterns and gives (6.18). At $x_* = 0.199$, the two fractions in that expression are less than 0.00327 and 0.01642. The inequality $(1+a)^{18} \leq (1-18a)^{-1}$, valid for $0 \leq 18a < 1$, proves $S_* < 0.081$ by rational arithmetic.

Since $o(d) \geq 3$, each $\eta_i x_*^2 \leq S_*/3$. Summing (6.17) over positions gives $\sum_i \eta_i \leq S_*/x_*^2$. Thus

$$\frac{\bar{\eta}}{1 - S_*/3} \leq \frac{0.081}{21(0.199)^2(1 - 0.081/3)} < 0.101.$$

□

For the moment fix N , and abbreviate $Z_i = Z_i^{(0,N)}$, $G_i = G_i^{(N)}$, and $x_i = |G_i|$. Define

$$h_i = 1 - \eta_i x_i^2, \quad F_i = h_i(1 + (1 - x_i^2)(Z_i G_i + \overline{Z_i G_i})), \quad i \in J_N. \quad (6.20)$$

These factors are bounded above and bounded away from zero uniformly in N, i, x_D . Indeed, $h_i \geq 1 - S_*/3$, and the other factor is at least $1 - 2x_* > 0$.

Proposition 6.7 (Bounded integral). *For every N and every fixed complex coordinate x_α ,*

$$\int_{\mathbb{R}} \psi(x_\beta) \prod_{i \in J_N} F_i(x_D) dx_\beta \leq \int_{\mathbb{R}} \psi(x_\beta) dx_\beta, \quad \psi(v) = \left(\frac{\sin(\pi v/10)}{\pi v/10} \right)^2, \quad (6.21)$$

where $\psi(0) = 1$.

Proof. We use the Fourier convention $e^{2\pi i \xi x_\beta}$. The nonnegative integrable function ψ has Fourier transform supported in $[-0.1, 0.1]$, hence in $[-1, 1]$. On suppressing the common frequency factor $-1/p'(\beta)$, the frequency of Z_i is β^i . Every phase appearing in G_i has frequency of absolute value at most β^{i-30} . For the type-1 term this follows from

$$|\beta^{i-33}(\beta^{-3} - 1 + \beta)| \leq \beta^{i-30}.$$

We call these predictor frequencies slow. Expanding a factor F_i gives monomials with one fast digit $d_i \in \{-1, 0, 1\}$, corresponding to $Z_i^{d_i}$, and at most five slow frequencies: two from h_i , two from $1 - |G_i|^2$, and one from G_i or its conjugate.

Expand the last block according to its fast digits, keeping all earlier factors for the moment. We first show that every inadmissible pattern has zero integral, even after multiplication by those earlier factors. Expand the latter and all slow pieces into monomials. If the current index is i , the fast sum accumulated from the top of the last block, divided by β^i , follows the recurrence $v \mapsto \beta v + d_i$. By (6.15), each allowed transition agrees with the state values (6.16) up to an injected error of absolute value at most β^{-30} . The propagated error, including at a first forbidden step, is bounded by

$$\frac{\beta^{-30} \beta^{21}}{\beta - 1} < 0.005. \quad (6.22)$$

At a first forbidden step, the ideal absolute value is at least $\beta(\beta - 1)$ for a departure from A_σ or B_σ , and at least 2 for a departure from C_σ . After (6.22), it is greater than 1.5. All lower fast digits together contribute at most $(\beta - 1)^{-1} < 1.192$ in the same units. If instead all transitions are allowed but the terminal state is A_σ or B_σ , the absolute value at the bottom index i is at least $\beta - 1 - 0.005$. The one-index gap then bounds the earlier fast terms by $1/(\beta(\beta - 1)) < 0.649$.

In either case, all slow terms, including those from earlier blocks, contribute at most

$$\frac{5\beta^{-30}\beta^{21}}{\beta-1} < 0.025 \quad (6.23)$$

relative to β^i . To see this, sum at most five frequencies of size β^{j-30} over indices j no larger than the top of the last block; that top is at most $i+20$. Thus the full frequency has absolute value greater than $0.1\beta^i/|p'(\beta)| > 1$. Its integral against ψ is zero.

After removing these zero integrals, the last block contributes $\prod_{i \in B_q} h_i$ times the sum over admissible fast patterns. The zero pattern contributes one. A nonzero pattern has absolute value at most $\prod_{d_i \neq 0} x_i$, since $0 \leq 1 - x_i^2 \leq 1$. The arithmetic-geometric mean inequality gives

$$\prod_{d_i \neq 0} x_i \leq x_*^{o(d)-2} \frac{1}{o(d)} \sum_{d_i \neq 0} x_i^2.$$

The absolute value of the retained pattern sum is therefore at most $1 + \sum_{i \in B_q} \eta_i x_i^2$. Writing $t_i = \eta_i x_i^2$, we have $0 \leq t_i < 1$ and

$$\prod_{i \in B_q} (1 - t_i) \left(1 + \sum_{i \in B_q} t_i \right) \leq 1;$$

for example, use $\prod (1 - t_i) \leq \exp(-\sum t_i)$. Since ψ and all earlier factors are nonnegative, removing the last block can only increase the upper bound for the integral. Repeat with the preceding blocks to obtain (6.21). $\square$

6.4 A typical exponential gain

We now evaluate the same factors in the stationary model, replacing Z_i, G_i in (6.20) by $\mathcal{Z}_i^{(0)}, \mathcal{G}_i$; denote the resulting factors by $\mathcal{F}_i$. The damping coefficients remain periodic with period 22, and are used at the same 21 positions per period.

Lemma 6.8 (A logarithmic estimate). *For $0 \leq x \leq x_*$ and $|u| \leq 2x$,*

$$\log(1 + (1 - x^2)u) \geq u - \frac{u^2}{2} - C_* x^3, \quad C_* := \frac{2}{3\sqrt{1 - x_*^2}}. \quad (6.24)$$

Proof. The assertion is immediate for $x = 0$. Otherwise let $D_x(u) = u - u^2/2 - \log(1 + (1 - x^2)u)$. Its derivative is

$$D'_x(u) = \frac{x^2 - x^2 u - (1 - x^2)u^2}{1 + (1 - x^2)u}.$$

For $u \geq 0$, this is at most the positive part of $x^2 - (1 - x^2)u^2$. Integrating that positive part on $[0, \infty)$ gives $2x^3/(3\sqrt{1 - x^2}) \leq C_* x^3$. On $[-2x, 0]$, the only possible interior critical point is a minimum, so the maximum is at an endpoint. We have $D_x(0) = 0$, whereas

$$\frac{d}{dx} D_x(-2x) = \frac{2x^2(1 - 2x - 4x^2)}{1 - 2x + 2x^3} \leq 2x^2.$$

As $D_0(0) = 0$, integration gives $D_x(-2x) \leq 2x^3/3 \leq C_* x^3$. $\square$

Proposition 6.9 (Growth on typical words). *For uniformly distributed words $a \in \{0, 1\}^N$,*

$$\Pr \left\{ \sum_{i \in J_N} \log F_i(x(a)) > 0.000220N \right\} \longrightarrow 1. \quad (6.25)$$

Proof. Apply Lemma 6.8 with $x = |\mathcal{G}_i|$ and $u = 2\mathcal{L}_i$. The identity

$$2\mathbb{E}\mathcal{L}_i^2 = \mathbb{E}|\mathcal{G}_i|^2 + \operatorname{Re} \mathbb{E}[(\mathcal{Z}_i^{(0)})^2 \mathcal{G}_i^2]$$

and the bound $\mathbb{E}|\mathcal{G}_i|^3 \leq x_* \mathbb{E}|\mathcal{G}_i|^2$ control the second and third order terms. Also

$$\log(1 - \eta_i |\mathcal{G}_i|^2) \geq -\frac{\eta_i |\mathcal{G}_i|^2}{1 - S_*/3}.$$

Set $\gamma = 22^{-1} \sum_{i \in B_0} \mathbb{E} \log \mathcal{F}_i$, where the division by 22 includes the unused index. Since $C_* x_* < 0.136$, Proposition 6.5 and Lemma 6.6 give the stationary mean rate γ with

$$\begin{aligned} \gamma &\geq \frac{21}{22} \left(2\mathbb{E}\mathcal{L}_0 - \left(1 + C_* x_* + \frac{\bar{\eta}}{1 - S_*/3} \right) \mathbb{E}|\mathcal{G}_0|^2 - \left| \mathbb{E}[(\mathcal{Z}_0^{(0)})^2 \mathcal{G}_0^2] \right| \right) \\ &> \frac{21}{22} (2(0.000210) - 1.237(0.000145) - 0.000007) > 0.000222. \end{aligned} \quad (6.26)$$

For completeness, we justify passage from this mean to finite words. Truncate each sequence C_A to $[-H, H]$, for a fixed integer H . The stationary logarithmic block sums formed from these truncated sequences depend on finitely many bits. They are uniformly bounded, have the same mean from block to block, and are independent when their blocks are sufficiently far apart. The variance of an average of Q such block sums is therefore $O_H(Q^{-1})$. Their averages converge in probability to their means.

Absolute summability of C_0, C_1 makes the truncated characters uniformly close to the original characters as $H \rightarrow \infty$. The predictors are finite sums with bounded total absolute weight, and the factors are uniformly bounded away from zero. Consequently the logarithms also converge uniformly under this truncation. Letting $H \rightarrow \infty$ after $N \rightarrow \infty$ proves

$$\frac{1}{N} \sum_{i \in J_N} \log \mathcal{F}_i \longrightarrow \gamma \quad \text{in probability.}$$

Couple the reversed word bits to $(b_0, \dots, b_{N-1})$ in this stationary sequence. For fixed H , the truncated finite and stationary characters used in a factor agree whenever $H + 2000 \leq i \leq N - 1 - H$. Only a number of boundary indices depending on H , not on N , remain. The same uniform approximation thus shows

$$\frac{1}{N} \left(\sum_{i \in J_N} \log F_i(x(a)) - \sum_{i \in J_N} \log \mathcal{F}_i \right) \longrightarrow 0$$

uniformly over the coupled bit sequences. Together with (6.26), this proves (6.25). $\square$

6.5 Counting typical vectors and proving singularity

The integral bound applies to continuous coordinates, whereas Proposition 6.9 concerns lattice points. The remaining step is to thicken each such point by a fixed radius. Normalization in the expanding directions makes the resulting change in the whole logarithmic product bounded independently of the word length.

Lemma 6.10 (Thickening word vectors). *There are constants $\rho > 0$ and $C_{\text{thick}} < \infty$, independent of N , such that the radius- ρ balls about distinct lattice points in Lemma 6.3 are disjoint, and, whenever y' lies in the ball about $y(a)$,*

$$\left| \sum_{i \in J_N} \log F_i(x'_D) - \sum_{i \in J_N} \log F_i(x_D(a)) \right| \leq C_{\text{thick}}, \quad x'_r = r^{-N} y'_r \quad (r \in D). \quad (6.27)$$

Proof. Choose $\rho \leq 1$ smaller than half the minimum distance between distinct points of the lattice. A coordinate change of size at most one in y changes the exponent of $Z_j^{(0,N)}$ by at most

$$2\pi \sum_{r \in D} \frac{|r|^{j-N}}{|p'(r)|}.$$

Every character occurring in F_i , including those in the type-1 term of the predictor, has index at most i . The logarithm of F_i is a uniformly Lipschitz function of these unit phases, since the coefficient sums and damping coefficients are bounded and F_i is uniformly positive. Its variation is therefore at most a fixed constant times $\sum_{r \in D} |r|^{i-N}$. Finally,

$$\sum_{i \in J_N} |r|^{i-N} \leq \sum_{i=0}^{N-1} |r|^{i-N} < \frac{1}{|r| - 1} \quad (r \in D).$$

This proves a bound independent of N . □

Proof of Theorem 6.1. The root assertion and the fact that β is not Pisot were proved above. Call a word typical if it satisfies the inequality in (6.25), and let $\mathcal{Y}_N$ be the set of distinct vectors $y(a)$ obtained from typical words. Typical words have probability tending to one. Lemma 6.10 implies that, for all sufficiently large N , the product $\prod_{i \in J_N} F_i$ is at least $\exp(0.000216N)$ throughout each of the disjoint balls about $\mathcal{Y}_N$.

All these balls lie in fixed bounded boxes in (x_D, y_I) coordinates, enlarged from those of Lemma 6.3. In the real expanding coordinate one may take $|x_\beta| \leq 3$. The function ψ has a positive minimum on $[-3, 3]$. Integrating first in x_β , Proposition 6.7 bounds the integral of the product over this fixed box by a constant independent of N . The product does not depend on y_I . Changing back to y coordinates contributes the Jacobian $\mathcal{M}^N$. Comparing with the disjoint balls, each of the same positive volume, gives

$$\#\mathcal{Y}_N \leq C_{\text{count}} \exp((\log \mathcal{M} - 0.000216)N) \quad (6.28)$$

for a constant C_{count} .

Let μ be the law of $\sum_{k \geq 0} b_k \beta^{-k}$ for independent fair bits. By reversal in (6.5), its length- N prefix has the distribution of $\beta x_\beta(a)$. Projecting $\mathcal{Y}_N$ to this coordinate cannot increase its cardinality. For each resulting prefix value, attach an interval of length $\beta^{-N}/(1 - \beta^{-1})$, which contains all possible remaining tails. Their union K_N is compact, $\mu(K_N) \rightarrow 1$, and (6.4) and (6.28) give

$$|K_N| \leq C_{\text{cover}} \exp((0.0002094 - 0.000216)N) = C_{\text{cover}} e^{-0.0000066N}. \quad (6.29)$$

Choose a subsequence N_j with $\mu(K_{N_j}) > 1 - 2^{-j}$. Then $\liminf_j K_{N_j}$ has full μ -measure and zero Lebesgue measure. Thus μ is singular. Finally,

$$\sum_{k \geq 0} (2b_k - 1)\beta^{-k} = 2 \sum_{k \geq 0} b_k \beta^{-k} - \frac{1}{1 - \beta^{-1}},$$

so its law $\nu_{1/\beta}$ is an affine image of μ and is singular as well. □

A A rational certificate for the degree-31 example

This appendix proves the root bounds in Proposition 6.2 and the predictor estimates in Proposition 6.5. The root calculation uses exact rational arithmetic. The moment calculation combines

finite rational products with an analytic estimate for their infinite tails. We specify the rounding and the order of operations so that the rational weights in (A.7) are completely determined. The accompanying script `verification/verify_degree31.py` implements these instructions using only Python integers and rational fractions; the recurrences below give the complete certificate independently of that implementation. Every terminating decimal in this appendix denotes its exact rational value.

A.1 Locating all the roots

Recall the polynomial

$$p(t) = \sum_{j=0}^9 (-t^{3j} + t^{3j+1}) - 2t^{30} + t^{31}.$$

Put $E = 10^{40}$. The entries in Table 1 are the real and imaginary numerators of centers $z = (a+ib)/E$. Include the conjugate of each nonreal center. Thus the table specifies 31 disks, all of radius $\delta = 10^{-35}$. Denote its first and sixth centers by z_β and z_α , respectively.

Real numerator	Imaginary numerator
18392867573155250211326543340571993487329	0
-9741384652837567528189111907762645564212	960611633124778624015947511546714789832
-9390979934023233037686839914141181659013	2848063934279278841200845525848576791934
-8707010393241200595697704637503228167256	4634358961359777463662031763152574795910
-772311855819973533585947045021501410451	6250214494152701297733561366714273855663
-6480873022754265587670654805348739312397	7617033927823246411701558316834470378693
-3296776282962928450195355562079974291919	9362810613239308448752821856735617305389
-1442513481645415489053007250852371013818	9733039088635544125591234293715446465839
473579005983151143865690217073943282984	9759329295153416086962240697595385836176
2372858204750359429049351486827380431084	9430908396110323024329804770889268443510
4184061662454447896993374181551299054853	8750376307352230997951514116951420713869
5840543823426662247726359012202878127980	7733999914049269380846159864123679516166
7280042164772721585016920411987150804420	6409668313572020921539025901166241650504
8444162922855313595050499086097221186662	4814843758334683329011653039104490553475
9275649862302447872891854755866108397375	2996533501299661135726051235425500195711
9715324892541617145821300300817663390044	1019107767839771580294306582422817927060

Table 1: Rational centers for the roots; both columns have denominator 10^{40} .

For a complex number write $\|z\|_1 = |\Re z| + |\Im z|$ and $\|z\|_\infty = \max(|\Re z|, |\Im z|)$. Exact Horner evaluation gives, at every displayed center,

$$|z| < 2, \quad \|p'(z)\|_\infty > 1, \quad 2 \cdot 10^{35} \|p(z)\|_1 < \|p'(z)\|_\infty. \quad (\text{A.1})$$

Here is an integer prescription for these comparisons. For $q(t) = \sum_{j=0}^d q_j t^j$ and the Gaussian integer $Z = Ez$, initialize $(V, h) = (q_d, 1)$ and, for $j = d-1, d-2, \dots, 0$, replace

$$h \leftarrow Eh, \quad V \leftarrow ZV + hq_j.$$

The final value is $q(z) = V/h$. Apply this to p and p' , and clear the positive denominators in (A.1). Comparisons of squared distances give $|z - z'| > .01$ for distinct centers, including conjugates. Comparisons of coordinates and squared moduli also give

$$\begin{aligned} 1.83928 < z_\beta < 1.83929, \quad 1.0002090 < |z_\alpha|^2 < 1.0002093, \\ |z| < .9939 \quad \text{at all other displayed centers.} \end{aligned} \quad (\text{A.2})$$

For $|h| = \delta$, the terms of degree at least two in $p(z+h) - p(z) - p'(z)h$ have total modulus at most

$$\delta^2 \sum_{j=0}^{31} |p_j| 3^j < 10^{17} \delta^2.$$

Indeed, $|z| < 2$ and $\delta < 1$, so the binomial expansion gives this bound term by term. By (A.1), the constant term is less than $|p'(z)|\delta/2$, whereas the displayed remainder is less than $|p'(z)|\delta/2$. Rouché's theorem therefore puts exactly one root, counted with multiplicity, in each disk. The disks are disjoint and their number is the degree, so these are all the roots and all are simple. The root in the real-centered disk is real by conjugation. Every other disk misses the real axis, so the remaining roots form fifteen nonreal conjugate pairs. The radius 10^{-35} and (A.2) imply every bound in Proposition 6.2, including $|1/\alpha| < .9999$.

A.2 Rounded coefficient arrays and finite products

The numerical inputs are the sequences C_0, C_1 defined in the main text in (6.6). We approximate their single and paired moments

$$P_A = \prod_{m \in \mathbb{Z}} \cos(\pi C_A(m)), \quad P_{AB}^\sigma(k) = \prod_{m \in \mathbb{Z}} \cos(\pi(C_A(m) + \sigma C_B(m+k))), \quad (\text{A.3})$$

where $A, B \in \{0, 1\}$, $\sigma \in \{1, -1\}$, and $k \geq 0$. Both products converge by exponential decay of the coefficients. The sign notation $+$ and $-$ means $\sigma = 1$ and $\sigma = -1$.

We now specify the approximations used to define the weights. Use

$$S = 10^{30}, \quad R = 3000, \quad H = 100, \quad K = 2000, \quad W = 10^9.$$

For a real rational x set

$$[x] = \frac{\lfloor Sx + 1/2 \rfloor}{S},$$

and round complex numbers componentwise. Thus, if $x = u/v$ with integers u, v and $v > 0$, its rounded numerator is $\lfloor (2Su + v)/(2v) \rfloor$. Additions, subtractions, conjugation, and multiplication by an integer are exact unless brackets are explicitly displayed. Set

$$\pi' = \frac{3141592653589793238462643383279}{S}.$$

The inequality $|\pi' - \pi| < 1/S$ follows by using 30 terms of the alternating series for $\arctan(1/5)$ and 10 for $\arctan(1/239)$ in Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$. The total remainder bound is

$$\frac{16}{61 \cdot 5^{61}} + \frac{4}{21 \cdot 239^{21}}.$$

In particular this verification, too, uses rational arithmetic only.

The contracting and expanding roots are denoted by I and D , as in the main text. For each representative center z first form $r_z = [1/p'(z)]$, with $p'(z)$ evaluated exactly. Store a base and a coefficient as follows:

$$(a_z, v_z) = \begin{cases} ([z], r_z), & |z| < 1, \\ ([1/z], -[1/z]r_z), & |z| > 1. \end{cases} \quad (\text{A.4})$$

For a contracting representative, start $v = v_z$ and add $2\Re v$ to $c_0(-1)$, then to $c_0(-2), c_0(-3), \dots$, replacing $v \leftarrow [va_z]$ after each addition. For an expanding representative do the same on $c_0(0), c_0(1), \dots$

the multiplier is 1 for the real root and 2 for the nonreal representative. All arrays start at zero. Compute $c_0(m)$ on

$$-R - K - 5 \leq m \leq H + K + 5$$

and define, wherever these three entries are present,

$$c_1(m) = c_0(m+3) - c_0(m) + c_0(m-1). \quad (\text{A.5})$$

These extra endpoint entries ensure that every subsequent use of c_1 lies in its defined range. The arrays approximate the sequences C_0, C_1 in the main text.

For a rational input c let $g(c), h(c)$ be the following rounded cosine and sine values. The displayed loop fixes the order of the summation.

```
x = [pi' c]; y = [x*x]
t = 1; u = x; g = t; h = u
for j = 1,...,50:
    t = [-t*y/(2*j*(2*j-1))]
    u = [-u*y/(2*j*(2*j+1))]
    g = g+t; h = h+u
return (g,h)
```

Write $g_A(m) = g(c_A(m))$ and $h_A(m) = h(c_A(m))$.

The only expanding root close to the unit circle is α and its conjugate. To treat their long tail, set

$$a = 1/\alpha, \quad u_0 = -a/p'(\alpha), \quad u_1 = u_0(a^3 - 1 + 1/a). \quad (\text{A.6})$$

Their contribution to $C_A(m)$ for $m \geq H$ is $u_A a^m + \overline{u_A} \overline{a^m}$. Let (a', u'_0) be the pair in (A.4) belonging to z_α . Compute

$$a'_0 = 1, \quad a'_{n+1} = [a' a'_n] \quad (0 \leq n < K), \quad u'_1 = [u'_0(a'_3 - 1 + [1/a'])].$$

The subscript in a'_n is a power-array index.

For a single moment use $w = u'_A$ and $k = 0$. For a paired moment of types $A, B \in \{0, 1\}$, sign $\sigma \in \{1, -1\}$, and lag $0 \leq k \leq K$, use $w = u'_A + \sigma[u'_B a'_k]$. In either case the following calculation returns an approximate moment. It retains the finite factors $-R - k \leq m < H$, omits the rapidly convergent negative tail, and treats the positive tail by summing the degree-2, 4, and 6 terms of $-\log \cos$ through geometric series. Their sum is the variable **pen**; the divisors 2, 12, 45 are the corresponding Taylor denominators. To compute $e^{-\text{pen}}$, the algorithm first approximates $e^{-\text{pen}/32}$ and then squares five times. This tail value is computed first, and the finite factors are then multiplied in increasing order of m . Section A.3 bounds all omissions and rounding errors.

```
U = [w*a'_H]
q_0 = 1; q_i = [q_{i-1}*U]          for i = 1,...,6
b_0 = 1; b_i = [b_{i-1}*pi']       for i = 1,...,6
pen = 0
for d = 2,4,6:
    mom = 0
    for j = 0,...,d:
        num = [q_j*conj(q_{d-j})]
        den = 1-[a'_j*conj(a'_{d-j})]
        mom = mom+binomial(d,j)*Re([num/den])
```

```

    pen = pen+[b_d*mom/f_d]          (f_2,f_4,f_6) = (2,12,45)
x = [-pen/32];  t = 1;  value = t
for j = 1,...,40:
    t = [t*x/j];  value = value+t
repeat five times: value = [value*value]
for m = -R-k,...,H-1, in increasing order:
    v = g_A(m)                      (single)
    v = [g_A(m)*g_B(m+k)-sigma*h_A(m)*h_B(m+k)] (pair)
    value = [value*v]
return value

```

Denote the outputs by $\tilde{P}_A$ and $\tilde{P}_{AB}^\sigma(k)$, respectively.

We use only lags $30 \leq k \leq 2000$. With $s = 0$ for $\sigma = 1$ and $s = 1$ for $\sigma = -1$, put $r = \tilde{P}_{00}^\sigma(k)$ and set

$$w_{ks} = \frac{\text{sgn}(r)}{W} \left\lfloor .68W \max(0, |r| - .00026) + \frac{1}{2} \right\rfloor, \quad 30 \leq k \leq 2000. \quad (\text{A.7})$$

Take $\text{sgn}(0) = 0$ and omit zero weights from subsequent sums. The recurrences above and this formula define the weights without any choices or numerical tolerances.

A.3 A uniform error bound for the moment calculation

We prove

$$|\tilde{P}_A - P_A| < \Delta, \quad |\tilde{P}_{AB}^\sigma(k) - P_{AB}^\sigma(k)| < \Delta, \quad \Delta = 5 \cdot 10^{-6}, \quad (\text{A.8})$$

for all types, signs, and lags used above. The important issue is the weakly contracting base $a = 1/\alpha$: truncating its powers at $H = 100$ would not be accurate. The geometric sums in the tail calculation retain all of these powers.

Coefficient and trigonometric errors. For $|z| < 2.001$, termwise differentiation gives

$$|p''(z)| < 4 \cdot 10^{13}.$$

Hence p' changes by less than $4 \cdot 10^{-22}$ between a center and its root. By (A.1), reciprocal differentiation and rounding in (A.4) give base errors less than $2/S$ and coefficient errors less than 10^{-21} . Direct squared modulus comparisons give $|a_z| < 1$ and $|v_z| \leq 1$ for every stored pair. The corresponding exact coefficients have modulus less than 1.001. Thus each recurrence step $v \leftarrow [va_z]$ propagates the existing error with factor at most 1 and adds less than $4/S$. There are fewer than 5100 such steps in any needed entry. Each coefficient-times-power term consequently has error less than $2 \cdot 10^{-21}$, and

$$|c_0(m) - C_0(m)| < 7 \cdot 10^{-20}, \quad |c_1(m) - C_1(m)| < 2.1 \cdot 10^{-19}. \quad (\text{A.9})$$

The finite recurrence also gives the following rational comparisons:

$$\begin{aligned} \max_{A,m} |c_A(m)| &< .847, & \max_{m \geq -20} |c_0(m)| &< .018, & \max_A |u'_A| &< .0087, \\ .8460 &< \max_{A,m} |c_A(m)| &< .8462. \end{aligned} \quad (\text{A.10})$$

The maxima here range over the computed entries; use squared moduli for the complex coefficients u'_A . To reproduce them, perform the coefficient recurrences in (A.4), the additions in (A.5), and the

displayed definition of u'_1 . These comparisons require neither evaluation of trigonometric functions nor a root-finding routine.

Every later trigonometric input is a sum of at most four of these coefficients, counted with multiplicity. Its error is less than 10^{-18} , and its rounded angle satisfies $|x| \leq 13$ and $|y| \leq 170$. The angle error is less than $4 \cdot 10^{-18}$. For the Taylor recurrences at a fixed value of y , each term-rounding error contributes at most e^{14}/S to the final sum: the subsequent factorial denominators bound the amplification by $\sum_{j \geq 0} 170^j/(2j)! < e^{14}$. Replacing x^2 by y costs at most $100e^{14}/S$, by differentiating the absolute term sums. The omitted Taylor tails at $|x| \leq 13$ are less than 10^{-30} . Together these effects give errors below 10^{-20} relative to the computed angle, and below 10^{-16} relative to the true cosine and sine. In particular every single or paired finite factor has error less than 10^{-15} .

There are at most $R + K + H = 5100$ finite factors. The true factors have modulus at most 1, and their approximations have modulus at most $1 + 10^{-15}$. Telescoping the product, including its roundings, shows that with an initial value of modulus at most 2 these errors contribute less than $3 \cdot 10^{-11}$.

Terms omitted at the two ends. On the negative side, $|C_0(m)| \leq 32(.994)^{-1-m}$ for $m < 0$, by summing the contracting-root terms. Including the three shifts for type 1, a single or paired coefficient at $m < -R - k$ is bounded by

$$200(.994)^{-4-(m+k)}.$$

Using $1 - \cos u \leq u^2/2$ and telescoping products, the total omitted product costs at most

$$\frac{3.142^2 200^2}{2} \frac{(.994)^{5994}}{1 - (.994)^2} < 5 \cdot 10^{-8}.$$

For example, the rational inequality $(.994)^{2996} < 3 \cdot 10^{-8}$ already suffices for this estimate. On the positive side the only terms omitted from the coefficients are those from β . Their bases are less than .545, and their type-0 and type-1 coefficients have modulus at most 4. The cosine Lipschitz bound therefore gives a total error at most $3.142 \cdot 8 \cdot (.545)^{100}/(1 - .545) < 10^{-20}$.

The geometric tail and its rounding. Let $w_* = u_A$ in the single case and $w_* = u_A + \sigma u_B a^k$ in the paired case. Put $U_* = w_* a^H$. The exact sums of the even powers of the weak-root tail are

$$\mu_d = \sum_{n \geq 0} (U_* a^n + \overline{U_* a^n})^d = \sum_{j=0}^d \binom{d}{j} \Re \frac{U_*^j \overline{U_*}^{d-j}}{1 - a^j \bar{a}^{d-j}} \quad (d = 2, 4, 6). \quad (\text{A.11})$$

The equality follows by the binomial theorem and absolutely convergent geometric series. In particular $\mu_d \geq 0$.

The power-array errors satisfy $|a'_n - a^n| \leq 3n/S$. Because $|a|, |a'| > .99$, the formula for u'_1 gives errors below $5 \cdot 10^{-21}$ for both u'_A , and $|u_A| \leq .00871$. It follows that the computed U has error below 10^{-19} ; its powers through degree 6 have errors below 10^{-18} . In each quotient in the algorithm, the numerator error is below $3 \cdot 10^{-18}$ and the denominator error is below 10^{-28} . The true and computed denominators have modulus greater than .00019: use

$$|1 - a^j \bar{a}^{d-j}| \geq 1 - |a|^d > 1 - (.9999)^2 > .00019$$

and the stated rounding error. Hence each quotient error is below $2 \cdot 10^{-14}$, the error in each computed μ_d is below $2 \cdot 10^{-12}$, and $|\mu_d| < 4 \cdot 10^5$ by the same denominator bound. The rounded powers of π'

through degree 6 have errors below 10^{-22} . Consequently **pen** approximates the nonnegative number

$$P = \frac{\pi^2 \mu_2}{2} + \frac{\pi^4 \mu_4}{12} + \frac{\pi^6 \mu_6}{45} \quad (\text{A.12})$$

to within 10^{-9} .

For $m \geq H$, the weak coefficient has angle bounded by

$$3.142 \cdot 4 \cdot .00871 \cdot (.9999)^m < .11(.9999)^m.$$

Summing the powers of this bound gives

$$P \leq \frac{.11^2}{2(1 - .9999^2)} + \frac{.11^4}{12(1 - .9999^4)} + \frac{.11^6}{45(1 - .9999^6)} < 31.$$

Thus $|\mathbf{pen}| < 32$ and $|x| < 1.1$ in the exponential calculation. The 40-term exponential recurrence has Taylor remainder less than 10^{-40} . Its rounding errors, bounded using $e^{1.1}$, are negligible compared with 10^{-9} . Each squaring amplifies its existing error by at most 3, since the true intermediate values for the computed penalty are at most $e^{10^{-9}}$. Including the error in (A.12), the returned exponential is within $4 \cdot 10^{-9}$ of e^{-P} .

The analytic remainder of the tail. For real z with $|z| \leq .11$ one has

$$\left| -\log \cos z - \frac{z^2}{2} - \frac{z^4}{12} - \frac{z^6}{45} \right| \leq .1z^8. \quad (\text{A.13})$$

Here are elementary bounds proving the stated constant. Set $y = z^2$, $h = 1 - \cos z$, and $h_0 = y/2 - y^2/24 + y^3/720$. Then $0 \leq h \leq y/2$ and $|h - h_0| \leq y^4/40320$. In $-\log(1 - h) = h + h^2/2 + h^3/3 + \dots$, the terms after the third are bounded by $h^4/(4(1 - h)) < .016y^4$. Substitution of h_0 in the first three terms costs less than $.00004y^4$. The terms through degree 3 are $y/2 + y^2/12 + y^3/45$; the sum of the absolute values of the remaining coefficients is less than .03, so their contribution is less than $.03y^4$ for $0 \leq y \leq .0121$. These bounds imply (A.13).

Applying (A.13) to all weak tail angles gives

$$\left| \sum_{m \geq H} -\log \cos(\pi(w_* a^m + \overline{w_* a^m})) - P \right| \leq \frac{.1(.11)^8}{1 - (.9999)^8} < 3 \cdot 10^{-6}.$$

Both penalties are nonnegative, so the same bound applies to the difference of their exponentials. Combining this with the rounding error, the finite-factor error, and the two omitted ends proves (A.8); indeed the sum of the stated errors is below $3.055 \cdot 10^{-6}$.

A.4 Correlation, variance, and the absolute Gram sum

For the rest of the appendix, abbreviate $Z_i = \mathcal{Z}_i^{(0)}$ and $G_i = \mathcal{G}_i$ from (6.9) and (6.11). All expectations are under the stationary law of the two-sided independent bits.

The preceding calculation defines exact rational weights and bounds each moment error uniformly. We next sum those moments to prove the first three estimates of Proposition 6.5. A final finite-product estimate will prove the fourth.

Write $X = -.001$, $Y = .0055$, and let $\mathcal{R}$ be the nonzero rows (w_{ks}, k, s) with $30 \leq k \leq 2000$. Define

$$b_A = \tilde{P}_A, \quad m_{k0} = \tilde{P}_{00}^+(k), \quad m_{k1} = \tilde{P}_{00}^-(k), \quad e = \tilde{P}_{01}^+(33).$$

The approximated correlation is

$$\tilde{c} = Xb_0 + Ye + \sum_{(w,k,s) \in \mathcal{R}} wm_{ks}. \quad (\text{A.14})$$

For each lag k , let J_{k0} and J_{k1} be respectively the second (minus) and first (plus) entries of the following pair:

$$\begin{cases} (\tilde{P}_{10}^+(k-33), \tilde{P}_{10}^-(k-33)), & k \geq 33, \\ (\tilde{P}_{01}^+(33-k), \tilde{P}_{01}^-(33-k)), & k < 33. \end{cases}$$

The approximated squared norm and absolute Gram sum are

$$\begin{aligned} \tilde{v} = & X^2 + Y^2 + 2XYb_1 + 2 \sum_{(w,k,s) \in \mathcal{R}} w(Xb_0 + YJ_{ks}) \\ & + \sum_{(w,k,s),(v,l,t) \in \mathcal{R}} wv m_{|k-l|, \mathbf{1}_{s=t}}, \end{aligned} \quad (\text{A.15})$$

$$\tilde{T} = \sum_{\substack{(w,k,s),(v,l,t) \in \mathcal{R} \\ k,l \geq 50}} |wv m_{|k-l|, \mathbf{1}_{s=t}}|. \quad (\text{A.16})$$

Both double sums are over ordered pairs. Equal signs require the minus moment in a conjugated phase pair; opposite signs require the plus moment. This explains the index $\mathbf{1}_{s=t}$, including the diagonal cases.

The exact finite calculations give

$$\begin{aligned} |\mathcal{R}| &= 753, \\ |X| + |Y| + \sum_{\mathcal{R}} |w| &= 198442946/W, \\ |X| + |Y| + \sum_{\substack{(w,k,s) \in \mathcal{R} \\ k < 50}} |w| &= 11816095/W, \\ WS\tilde{c} &= 211815537786494774284357596751188225, \\ W^2S\tilde{v} &= 143920884250294630908494564120361905955993456, \\ W^2S\tilde{T} &= 105303788239450118397416374596041413014338334. \end{aligned} \quad (\text{A.17})$$

These are integer equalities: compute the arrays and products in Section A.2, round the weights by (A.7), and use the unrounded rational sums (A.14)–(A.16). In particular

$$211810 < 10^9 \tilde{c} < 211820, \quad 143910 < 10^9 \tilde{v} < 143930, \quad 105300 < 10^9 \tilde{T} < 105310.$$

This gives $|G_i| < .199$ and $|G_i^{(N)}(x)| < .199$ by the triangle inequality. The total correlation error is at most $\Delta \cdot .199$, and the squared-norm error is at most $\Delta \cdot .199^2$. Thus

$$\mathbb{E}\Re(Z_i G_i) > .0002108205 > .000210, \quad \mathbb{E}|G_i|^2 < .000144119 < .000145.$$

To prepare for the last estimate, call the constant row, the type-1 row, and the rows with $k < 50$ the *small rows*; the remaining rows are the *big rows*. A row of weight w_v has phase $U_v = \exp(\pi i \sum_m f_v(m) e_m)$, where $e_m = 2b_{i+m} - 1$. Its sequence f_v is zero for the constant, $C_1(m+33)$ for the type-1 row, and $(1-2s)C_0(m+k)$ for a row (w, k, s) . Define the true absolute Gram sum

$$T = \sum_{v, v' \text{ big}} |w_v w_{v'} \mathbb{E} U_v \overline{U_{v'}}|. \quad (\text{A.18})$$

The error between this sum and (A.16) is at most $\Delta \cdot .199^2$, by $||x| - |y|| \leq |x - y|$. Hence

$$T < .000116, \quad \sqrt{T} < .011. \quad (\text{A.19})$$

The total absolute weight of the small rows is less than .012.

A.5 Finite products controlling the remaining moment

We now bound $\mathbb{E}(Z_i^2 G_i^2)$. The big rows have small coefficients on the 71 positions $-70 \leq m \leq 0$. More precisely, with $t = .06$, (A.9) and (A.10) give

$$|\tan(\pi f_v(m))| \leq t \quad (v \text{ big}, -70 \leq m \leq 0). \quad (\text{A.20})$$

Indeed $m + k \geq -20$ for these rows, and the coefficient bound .018, together with $\sin u \leq u$ and $\cos u \geq 1 - u^2/2$, is sufficient.

For any real sequence V put $c_m = |\cos(\pi V(m))|$ and $s_m = |\sin(\pi V(m))|$. The finite products needed below are

$$\begin{aligned} B_2(V) &= \prod_{m=-70}^0 \min \left(1, \frac{(1+t^2)c_m + 2ts_m}{1-t^2} \right), \\ B_1(V) &= \prod_{m=-70}^0 \min \left(1, \frac{c_m + ts_m}{\sqrt{1-t^2}} \right), \\ B_0(V) &= \prod_{m=-200}^0 c_m. \end{aligned} \quad (\text{A.21})$$

Their bounds are

$$B_2(2C_0) < .011, \quad \max_{f \text{ small}} B_1(2C_0 + f) < .015, \quad \max_{f,g \text{ small}} B_0(2C_0 + f + g) < .0032. \quad (\text{A.22})$$

Here the maxima range over the actual phase sequences of the small rows. Besides the constant and the type-1 row, their eight nonzero rows are as follows:

k	32	36	37	39	40	43	44	47
s	0	1	0	1	0	0	1	1
Ww_{ks}	-536932	1265024	331701	47119	1483338	436861	458216	756904

For completeness, the following upper-rounding calculation verifies all of (A.22). For each of its three cases and each indicated choice of small rows, form the approximate sequence V' by replacing C_A by c_A . Initialize $v = 1$. For increasing m in the relevant interval, compute

$$\begin{aligned} b &= |g(V'(m))| + 10^{-15}, & y &= |h(V'(m))| + 10^{-15}, \\ F_0 &= b, & F_1 &= S^{-1} + [1.002(b + .06y)], \\ & & F_2 &= S^{-1} + [1.004(1.0036b + .12y)], \\ v &\leftarrow S^{-1} + [v \min(1, F_j)] & & \text{in case } j. \end{aligned}$$

The factors 1.002 and 1.004 bound $(1-t^2)^{-1/2}$ and $(1-t^2)^{-1}$, respectively. The added 10^{-15} exceeds the trigonometric error proved above, and each added S^{-1} compensates for a possibly downward

rounding. Thus this calculation gives upper bounds, including for B_0 because its true factors are at most 1. The maximum final numerators at scale S , in the order B_2, B_1, B_0 , are

$$\begin{aligned} &10861284406435197401851134503, \\ &14731866010915969694738277129, \\ &3155364376927094122833206055. \end{aligned} \tag{A.23}$$

Multiplication by $10^6/S$ bounds them respectively by 10862, 14732, and 3156, proving (A.22).

A.6 From the finite products to the anisotropy bound

The estimates just obtained concern individual coefficients and finite products. We finish by showing how they control the full sum of phases without expanding every three-phase moment. Write $G_i = G_{\text{small}} + G_{\text{big}}$ according to the preceding partition, and fix any subset $J \subset \{-70, \dots, 0\}$. Expanding the big phases in the independent signs e_m on J gives

$$G_{\text{big}} = \sum_{A \subset J} i^{|A|} e_A H_A, \quad e_A = \prod_{m \in A} e_m, \tag{A.24}$$

where

$$H_A = \sum_{v \text{ big}} w_v U_v^{\text{out}} \prod_{m \in J} \cos(\pi f_v(m)) \prod_{m \in A} \tan(\pi f_v(m)).$$

The phase U_v^{out} depends only on signs outside J , so each H_A is independent of the signs on J .

We claim that

$$\|H_A\|_2 \leq \sqrt{T} t^{|A|} (1 - t^2)^{-|J|/2}. \tag{A.25}$$

For two big rows, independence gives their full Gram entry as the outside Gram entry times

$$\prod_{m \in J} \cos(\pi f_v(m)) \cos(\pi f_{v'}(m)) (1 + \tan(\pi f_v(m)) \tan(\pi f_{v'}(m))).$$

By (A.20), each last factor is at least $1 - t^2$. In the expansion of $\|H_A\|_2^2$, the extra tangent products have modulus at most $t^{2|A|}$. Taking absolute values term by term therefore bounds the squared norm by $T t^{2|A|} (1 - t^2)^{-|J|}$, proving (A.25).

First consider $\mathbb{E}(Z_i^2 G_{\text{big}}^2)$ and set $V = 2C_0$. For a pair $A, A' \subset J$ in (A.24), averaging the signs on J contributes a factor of modulus c_m where membership in A, A' agrees, and a factor of modulus s_m where it differs. The outside expectation is bounded by $\|H_A\|_2 \|H_{A'}\|_2$ using Cauchy–Schwarz and the unit modulus of the outside part of Z_i^2 . Summing over the four membership choices at each site gives

$$|\mathbb{E}(Z_i^2 G_{\text{big}}^2)| \leq T \prod_{m \in J} \frac{(1 + t^2)c_m + 2ts_m}{1 - t^2}.$$

Choose J to contain precisely those positions whose displayed factor is less than 1. This yields

$$|\mathbb{E}(Z_i^2 G_{\text{big}}^2)| \leq T B_2(2C_0). \tag{A.26}$$

For a fixed small phase U with sequence f , use the same expansion with $V = 2C_0 + f$. There is now one subset A ; summing its two membership choices produces $c_m + ts_m$. The outside expectation is bounded by $\|H_A\|_2$. Optimizing J as before gives

$$|\mathbb{E}(Z_i^2 U G_{\text{big}})| \leq \sqrt{T} B_1(2C_0 + f).$$

For two small phases with sequences f, g , independence directly bounds their full moment by $B_0(2C_0 + f + g)$: discard all the other absolute cosine factors, which are at most 1.

Finally sum these bounds with the absolute row weights. Using (A.19), (A.22), and the small-weight bound .012, we obtain

$$\begin{aligned} |\mathbb{E}(Z_i^2 G_i^2)| &\leq T \cdot .011 + 2(.012)\sqrt{T} \cdot .015 + (.012)^2 \cdot .0032 \\ &< .000116 \cdot .011 + 2 \cdot .012 \cdot .011 \cdot .015 + .012^2 \cdot .0032 \\ &= .0000056968 < .000007. \end{aligned}$$

This proves the remaining estimate of Proposition 6.5.

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